HomeoClinic MVP1 Architecture Design

Open `index.html` (fully static, works from disk) for the review version with all diagrams inline.

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01-legacy-analysis.md	What the existing 2018 Doctor Clinic app is, its data model, its problems, what to keep
02-architecture-overview.md	Context, containers, components, threading, deployment, security, ADRs
03-repertory-engine-design.md	Rubric search, scoring engine, pack format, content pipeline, citations, UI
04-licensing-payments-design.md	MVP1 offline UPI QR + WhatsApp + emailed signed licence flow, licence formats, identity binding, Vendor Licence Tool, states; MVP2 backend and hosted UPI
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Diagrams: sources in `diagrams/src/*.puml`, rendered in `diagrams/svg/`. Rebuild everything with `./build.sh`.

Requirements: `../requirements/` (copy of the approved requirement set).

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01. Legacy "Doctor Clinic" (2018) Analysis

Source: `~/Doctor-Clinic/DoctorClinic/` (jar dated Dec 2017, logs to Dec 2018). Only compiled classes are present; there is no Java source except one Eclipse WindowBuilder helper. Findings come from the jar class list, the MySQL schema (`Original Database/doctorclinic.sql`), Jasper report templates, folder layout and the error logs. Patient photos, PDFs and the SQL dumps contain real patient data and were not opened or copied.

Diagram: diagram 13-legacy-architecture.

1. What it is

Aspect	Legacy implementation
UI	Java Swing, JTattoo look and feel, ~66 screen/helper classes in the default package (Home, LoginPage, PatientDetailsPanel, CasePaper, FollowUp, GeneralPrescription, CustomerPayment, ViewAppointMent, BirthdayReminder, ReferLetter, …)
Database	Separate MySQL server over JDBC (`DBConnect` , `SettingDatabase`), latin1 charset, 19 tables
Printing	JasperReports (17 `.jrxml` templates) + iText: case paper, follow-up, prescription, receipt, certificate, envelope, instruction, diet, declaration, referral letter
Reports	Daily collection, date-wise payment/ledger/follow-up, medicine report, Excel export via jxl
Photos	Webcam capture via OpenCV (`ImageCapture`), files in `Patient Photos/`
Local language	Marathi instructions printed using legacy Shivaji fonts (non-Unicode font hack)
Licence	`KeyGen` , `Encryptor` , `RenewalLicense` , `LicenseExpired` ; `tbllicenceinfo(EndDate)` single row in the database; `GetComputerName` hints at machine binding
Backup	Manual `.sql` dump to `Backup/`
Packaging	Single 33 MB fat jar; needs a Java runtime and a MySQL server installed and running

2. Legacy data model (19 tables)

Group	Tables	Notes
Patient	`tblpatient` (CaseNo, name, address, contact, sex, dob, age, marital, education, referred by, occupation, weight, height)	Composite PK (Id, CaseNo); no mobile normalisation, no photo column (photo by file name)
Case	`tblchiefcomplaint` (Complaint1..3), `tblhistory` (past, family, desires, aversions, thirst, stools, thermals, urine, perspiration, sleep), `tblgeneralities` (sun, fasting, sleep loss, getting wet, ascending, BP, dream, pulse), `tblmentals` , `tblmenstrualhistory`	One row per case per table, fixed columns, free text
Visits	`tblfollowup` (date, weight, complaints, diagnosis, medicine, rubrics, charges, status)	Rubrics and medicines stored as free text, not linked to any repertory
Rx	`tblpriscription`	Copies patient name/address/age into the Rx row (denormalised), medicine as text

Group	Tables	Notes
Money	`paymentcustomer` (total, paid, balance, now paid, mode, flags), `tblreceipt`	Money as `double` (rounding errors)
Other	`tblappointment`, `tbldiagnosis`, `tbldiet`, `tblinstruction` (with language), `tbldeclaration`, `tblreports` (file path), `tblclinicsetting`, `login`, `tbllicenceinfo`	Single clinic, single user

3. Problems seen (what the new design must fix)

#	Problem	Evidence	New design response
L1	Needs a MySQL server; connection drops	13 CommunicationsException, 11 connection-closed errors in logs	Embedded SQLite, no server (ADR-002)
L2	SQL errors at runtime	53 MySQLSyntaxErrorException, 148 SQLException traces; likely string-built SQL (also an SQL injection risk)	Parameterised queries only, compile-time checked SQL in Rust
L3	No repertory at all; rubrics are typed as text	`tblfollowup.Rubrics` text column	Integrated repertory engine (doc 03)
L4	Only 3 complaints, fixed case fields	`Complaint1..3`, fixed columns	Template-driven case JSON (CASE-004)
L5	Licence is one editable date in the DB	`tbllicenceinfo(EndDate)`	Signed Ed25519 licence file (doc 04)
L6	Plain-text or weakly protected passwords and no data encryption	`login.UserPassword text`	Argon2id PIN hash, SQLCipher DB
L7	Non-Unicode Marathi via special fonts	Shivaji*.ttf, latin1 DB	UTF-8 everywhere, Noto Devanagari fonts
L8	Money stored as `double`	`paymentcustomer`	Integer paise (`INTEGER`)
L9	Manual SQL dump backups only	`Backup/backup.sql`	Automatic encrypted rotating backups (BAK)
L10	Heavy install (Java + MySQL + 33 MB jar)	Folder layout	Single small installer, no runtime prerequisites

4. Features to keep (doctors already rely on them)

Case paper print, follow-up history, general prescription, payment receipt with amount in words (`NumtoWord`), daily collection and date-wise reports, appointments, birthday reminders, diet and instruction sheets in Marathi, medical certificate, referral letter, envelope print, declaration print, patient report uploads, webcam photo, theme setting.

Gap against current requirements: certificate, referral letter, envelope, declaration and birthday reminder are not in 01-mvp1-requirements.md. Proposed as new IDs in `06-review-questions.md` (Q1).

5. Migration path (MVP5, de-prioritised)

Migration is built only after MVP1-MVP4 work end to end (requirement IMP-002). The design below is kept ready.

A one-time importer (`tools/legacy-import`, see 05-data-design.md section 6) reads either a live legacy MySQL database or a `.sql` dump and maps all 19 tables into the new schema, so existing users can switch without losing records.

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02. MVP1 Architecture Overview

Scope: MVP1 from `../requirements/`: a complete offline desktop app for Windows and Linux with no hosted servers. Doctors pay to our business UPI ID through a QR the app generates offline, send proof on WhatsApp, and receive a signed licence by email from our offline Vendor Licence Tool (requirements 03 section 3, design 04). The hosted backend shown in the diagrams is MVP2 (built when MVP1 revenue funds it); it is designed now so MVP1 choices do not block it. Migration is MVP5.

1. System context

Diagram: [diagram 01-system-context]

Actor / system	Interaction with MVP1
Doctor	Uses the desktop app all day, fully offline
Receptionist	Uses the same PC (separate login only from MVP2)
Patient	Receives printed or WhatsApp-shared Rx and receipts; pays the doctor by the doctor's own UPI QR
HomeoClinic Desktop	The product. Everything clinical runs here
Vendor Licence Tool (MVP1)	Internal offline tool; operator issues signed licence codes

Actor / system	Interaction with MVP1
Licence & Payment Service (MVP2)	Our hosted backend. Issues signed licences automatically; never receives patient data
Business UPI ID / bank (MVP1)	Receives payments from the offline Pay QR; dashboard shows UTRs for verification
WhatsApp Business (MVP1)	Doctor sends payment proof via pre-typed click-to-chat QR
Email (MVP1)	Licence file, certificate PDF and invoice to the doctor
Razorpay (MVP2)	Orders, webhooks, AutoPay
SMS / Email provider (MVP2)	Sends licence keys and GST invoices automatically
WhatsApp (wa.me link)	Opened on the doctor's PC for sharing a Rx; no API in MVP1
USB drive / folder	Backup target, offline update files, content packs

2. Containers

Diagram: diagram 02-containers

Container	Tech	Responsibility
Desktop UI	React 18 + TypeScript in Tauri webview	Screens, forms, keyboard flows, print preview
Desktop Core	Rust (Tauri 2 backend)	All business logic, storage, search, repertory engine, licence check, backup, PDF
clinic.db	SQLite + SQLCipher	Patients, visits, Rx, bills, audit, settings
user_content.db	SQLite + SQLCipher	Doctor's repertory additions, citations, bookmarks, synonyms
content/*.hcpack	Signed read-only SQLite packs	Repertories, Materia Medica, synonyms, relationships
files/	Encrypted blobs	Photos, attachments, generated PDFs
Vendor Licence Tool (MVP1)	Rust + Tauri, offline, hardware signing key	Verify UTR, sign licences, certificate PDF, invoice, register, register, GST invoices
Licence API (MVP2)	Rust axum, stateless	Activate, offline activate, deactivate, refresh, webhook
Portal web (MVP2)	React (mobile first)	Buy, renew, manage devices, invoices

Container	Tech	Responsibility
Licence DB (MVP2)	PostgreSQL	Customers, orders, licences, devices, invoices
Signing key	MVP1: hardware key (YubiKey/smart card); MVP2: Cloud KMS (ap-south-1)	Ed25519 private key; never leaves KMS

3. Desktop component design

Diagram: diagram 03-desktop-components

Rust workspace crates (hexagonal style: domain has no IO, adapters at the edges):

Crate	Contains	Depends on
`hc-domain`	Entities, value types (Money in paise, PatientCode, Potency), validation rules	nothing
`hc-storage`	SQLCipher pool (1 writer + N readers), migrations, repositories, FTS triggers, backup/restore	domain
`hc-search`	Patient search query builder, phonetic keys, ranking	storage
`hc-repertory`	Pack loader/verifier, rubric search, bitset scoring engine, analysis methods, citations merge	domain
`hc-licence`	Licence parse/verify, identity lock, clock guard, entitlements, offline UPI and WhatsApp QR, offer codes	domain (shared with Vendor Licence Tool and MVP2 server)
`hc-docs`	Print/PDF rendering from HTML templates, amount in words, Indian number format	domain
`hc-app`	Tauri commands (the only API the UI can call), app services orchestrating use cases, event bus to UI	all above

UI modules mirror use cases: `home/today`, `patients`, `case`, `visit`, `repertory`, `materia-medica`, `prescription`, `billing`, `reports`, `backup`, `settings`, `licence`.

Rules: - UI never sees SQL or file paths; it calls typed commands generated with `tauri-specta`. - Every write is a single use-case command in one DB transaction, with an audit row. - Long work (search, analysis, backup, pack install) runs on worker threads; UI gets progress events and can cancel.

4. Runtime and threading

Thread / pool	Work
UI (webview)	Rendering only
Tauri async runtime (tokio)	Command dispatch
DB writer (1 dedicated connection)	All writes, serialised, WAL mode
DB readers (4 connections)	Searches, lists, reports; run in parallel with writes
Repertory engine	Pack data memory-mapped; rubric bitsets cached in an LRU (64 MB cap)
Background jobs	Auto-backup, index optimise, periodic licence re-check (local), optional update check (signed static manifest); MVP2 adds online licence refresh

Startup sequence: diagram 14-seq-startup. Target cold start < 4 s (NFR-001): unlock DB, verify licence, open packs lazily (only metadata at start), render Today screen.

5. Deployment

Diagram: diagram 04-deployment

Target	Artifact	Install location	Data location
Windows 10/11 x64	Signed NSIS `.exe` and `.msi`, offline WebView2 bundled	`%LOCALAPPDATA%\Programs\HomeoClinic` (per-user, no admin)	`%APPDATA%\HomeoClinic\`
Linux x64	AppImage and `.deb`	`/opt/homeoclinic` or AppImage anywhere	`~/.local/share/homeoclinic/`
Backend (MVP2)	Containers on AWS ap-south-1 (ECS Fargate or a small VM), RDS PostgreSQL, KMS		India region only

Data folder layout:

```
HomeoClinic/
  clinic.db  user_content.db  settings.json(non-secret)
  content/   kent-1.0.hcpack  boericke-1.0.hcpack ...
  files/     <uuid>.bin (encrypted)
  backups/   daily/ weekly/ monthly/
  logs/      app.log (no patient data)
  licence/   licence.json  state.bin
```

6. Security architecture

Diagram: key hierarchy in diagram 12-state-licence (licence) and diagram 11-seq-backup-restore (backup keys).

Concern	Design
At rest	Data Encryption Key (DEK, random 256-bit) encrypts clinic.db, user_content.db (SQLCipher) and files (AES-256-GCM). DEK is wrapped by a Key Encryption Key derived from the PIN (Argon2id) and also by the OS store (Windows DPAPI / Linux Secret Service). Recovery code wraps a third copy.
Backups	Separate backup key from backup password (Argon2id); `.hcbak` = header + encrypted tar of DB snapshot and files
Network	MVP1 desktop opens no listening ports and has no server to call; it only hands wa.me / mailto links to the OS and optionally reads a signed update manifest over TLS 1.2+. MVP2: licence API with certificate pinning
Packs and updates	Ed25519 signatures verified before use; two embedded public keys for rotation
Backend (MVP2)	Auth on every non-public endpoint, admin behind SSO + 2FA, webhook HMAC check, rate limits, KMS signing, no patient data
Code	Parameterised SQL only, `cargo audit` / `npm audit` in CI, pinned dependencies, Content Security Policy in webview, Tauri capability allowlist (no shell, no arbitrary FS)
Tamper / crack protection (TAM-001..013)	Layered, see section 6b

6b. Tamper-proof release build

Diagram: diagram 18-secure-release-pipeline.

No client binary is truly uncrackable, so protection is layered so that cracking is expensive and a cracked copy is still of little value:

Layer	What it does	Requirement
1. Single native blob	Rust compiles to machine code (not decompilable like the legacy Java jar). UI assets are embedded in the exe, encrypted at build time, decrypted in memory only	TAM-001
2. Hardened compile	Fat LTO, stripped symbols, no debug info, devtools and debug commands compiled out, minified/mangled frontend	TAM-002
3. Hidden secrets	Compile-time string encryption for keys, URLs, licence messages	TAM-003
4. Many licence checks	Licence logic inlined at startup, new-visit/Rx/bill, background timer (MVP2: also pack open); each verifies differently	TAM-004
5. Self-integrity	Signed hash manifest of code sections embedded at build; randomised runtime re-hash; Authenticode self-check on Windows	TAM-005, TAM-006
6. Anti-debug / anti-hook	Debugger and injection detection; silently disables paid paths	TAM-007
7. Targeted obfuscation	Control-flow obfuscation only on `hc-licence` and pack-decrypt code	TAM-008
8. Server-held value (MVP2)	Paid packs encrypted, key only inside device-bound licence; updates, cloud, sync, SMS require server-checked licence	TAM-009, LIC-031
9. Traceability	Per-licence watermark and identity printed on every document (MVP1, TAM-014); server abuse detection and revocation (MVP2)	TAM-010, TAM-011, TAM-014

Release pipeline (TAM-012): build on a locked CI runner -> embed and encrypt assets -> compile hardened -> obfuscate licence modules -> compute and sign code-section manifest -> inject manifest -> Authenticode/GPG sign via hardware key / HSM (MVP2: KMS) -> automated release scan (no symbols, no devtools, no plain secrets, signatures valid, Defender scan clean) -> publish. Any failed step blocks the release.

Response policy (TAM-013): tamper detection only blocks new paid actions (Read-only mode). Patient data is never deleted, encrypted or hidden, and the PC is never affected.

UX guardrails (SUX-001..012): every layer above is designed to be invisible to a genuine doctor.

Layer	How it stays invisible
Encrypted embedded assets	Decrypted once at startup into memory (about 20 MB, < 150 ms), counted in the 4 s cold-start budget

Layer	How it stays invisible
Obfuscation	Applied only to `hc-licence` and pack-decrypt, never to search, scoring, DB or UI code, so hot paths keep full speed
Licence checks	Run in background or piggyback on write commands (< 1 ms each); no dialogs
Self-integrity	Background idle-time hashing in < 5 ms chunks; restriction needs 2 independent failures; first response is automatic Repair
Anti-debug	Silent, only affects paid paths, never crashes or shows warnings
Paid pack keys (MVP2)	Unwrapped once per session at first pack open (< 50 ms)
Signing	EV certificate so no SmartScreen warning; per-user install, no admin, no firewall prompt
Restriction	Never interrupts an ongoing visit (SUX-008); friendly message with one action button (SUX-007)

Verification: performance and usability tests run on the fully protected release build (SUX-001, acceptance flow 13).

Not used: kernel drivers / rootkit-style DRM and whole-binary packers (UPX, VMProtect) because they cause antivirus false positives and erode doctor trust.

7. Architecture decisions (ADR summary)

ADR	Decision	Alternatives rejected	Reason
001	Tauri 2 + React	Electron, Flutter, Java Swing (legacy), .NET	Small installer, low RAM, Win+Linux+mobile from one codebase
002	Embedded SQLite (SQLCipher)	MySQL/PostgreSQL local server (legacy)	No server to install or crash (legacy L1); single-file backup
003	Rust core with all logic	Logic in TypeScript	Performance for search/scoring, memory safety, shared licence crate with server
004	Content as signed read-only packs	Content in clinic.db	Add repertories without app updates (PACK-006); patient data untouched by content updates
005	Case data as template-driven JSON in `visit.data_json` + FTS	Fixed columns per field (legacy)	Templates are doctor-editable (CASE-004)

ADR	Decision	Alternatives rejected	Reason
006	Bitset scoring engine	SQL GROUP BY scoring	< 300 ms recompute (NFR-004) on low-end PCs
007	Ed25519 licence file + QR offline activation	DB date (legacy), online-only check	Works offline, tamper-proof
008	HTML templates rendered to PDF by the webview	JasperReports (legacy), native PDF lib	Doctor-editable layouts, Unicode Indian scripts
009	Money as integer paise	double (legacy)	No rounding errors
010	UUIDv7 primary keys + `row_version`	Auto-increment	Ready for MVP2 LAN and MVP3 sync without key clashes
011	Layered anti-tamper: hardened single native binary + integrity checks + server-held value (encrypted paid packs)	Whole-binary packer/VMProtect, kernel DRM, no protection	Strong enough to make cracking uneconomic, no antivirus false positives, never harms patient data

8. Cross-cutting

- Logging: structured JSON logs, patient identifiers never logged (IDs hashed), 10 MB x 5 rotation.
- Errors: domain errors mapped to friendly messages with a suggested action (NFR-033).
- i18n: all UI strings in `locales/en.json` from day one; Noto Sans / Noto Sans Devanagari bundled for print.
- Accessibility: keyboard-first, focus rings, min font 14 px, WCAG AA contrast.
- Testing: unit tests per crate, golden tests for scoring and search ranking, Playwright E2E for the 11 core flows, perf tests with 50k/100k seeded data.

9. How MVP1 prepares for MVP2/MVP3

Later need	MVP1 hook
LAN multi-user	All UI calls go through `hc-app` commands, which can be exposed over an authenticated LAN API later without UI change
Cloud backup	`.hcbak` format and backup key already end-to-end encrypted
Mobile sync	UUIDv7 keys, `row_version`, `updated_at`, append-only visits, audit log as change feed
Roles	`user` table and `created_by/updated_by` exist from MVP1
Languages	i18n strings and synonym `lang` column exist

03. Repertory Engine Design

Requirements: REP-, *MM-*, PACK-, *CIT-* in `requirements/02-repertory-engine.md`. Diagrams: diagram 06-er-content, diagram 08-seq-repertorise, diagram 15-pack-pipeline.

1. Components

Component	Responsibility
PackRegistry	Discovers `content/*.hcpack`, verifies signature + checksum + entitlement, attaches each pack DB read-only, exposes enabled packs
RemedyMaster	Merged remedy list across packs (MM-001, PACK-008); maps pack remedy codes to global remedy ids
RubricSearch	FTS5 trigram over `full_path` + synonym expansion + abbreviation expansion; ranking
RubricStore	Rubric tree navigation, rubric detail, cross-refs; reads pack DB + `user_content.db` overlay
ScoringEngine	Builds per-rubric remedy bitsets and grade arrays; runs analysis methods
CitationService	CRUD for sources/citations/links; overlay of doctor additions
MMReader	Materia Medica sections, full-text search, compare

2. Rubric search pipeline (REP-010..017)

1. Normalise input: lowercase, strip punctuation, British/US spelling map.
2. Expand: each word -> {word, synonyms(word, lang), abbreviation forms}; multi-word synonyms matched first ("loose motions" -> "diarrhoea").
3. Query every enabled pack's `rubric_fts` with `MATCH` on expanded terms (AND across user words, OR within a word's expansions).
4. Rank: score = BM25 + 3 x (all words in leaf text) + 2 x (chapter matches a chapter synonym) + 1 x log(remedy_count) - 0.5 x depth. Top 50.
5. Return rows with full path, remedy count, top 5 remedies (precomputed column `top5_json` in pack).

Budget: < 100 ms per keystroke (NFR-003). Query cancelled when the next keystroke arrives.

3. Scoring engine (REP-033..036)

Data per analysis (built when a rubric is added, cached by rubric id): - `present: BitSet` over global remedy index (about 2,500 remedies = 40 x u64) - `grade: [u8; N]` sparse map for remedies present

(normalised 1..3 via pack `grade_scale`) - doctor additions from `user_rubric_remedy` merged when "include my additions" is on (CIT-008), marked with a flag bitset

Algorithm (all methods share one loop):

```
candidates = ALL
for r in eliminative rubrics: candidates &= present[r]
for m in candidates:
    for r in active rubrics:
        g = grade[r][m]; if g == 0: continue
        cover[m] += 1
        score[m] += g * w(r) * chapterWeight(method, r.chapter)
        if method.smallBonus and r.remedy_count < T: score[m] += B
        if method == Boenninghausen: add parent rubric grades at g-1
sort by method key (score, cover, abbrev)
```

Complexity: 30 rubrics x 2,500 remedies = 75k operations, well under 1 ms. The 300 ms budget is for the UI.

Snapshot (REP-038): rubric list + settings + top 20 results stored in `analysis.result_snapshot_json` so an old analysis opens identically even after packs change.

4. Pack format (PACK-001..013)

```
eval-publicum-1.0.hcpack  (zip, stored, not compressed for mmap)
  manifest.json      { pack_id, type, title, edition, version, format_version, grade_max,
                       language, distribution: public|internal_eval, licence_spdx,
                       source_ref (registry row), licence: free|sku, min_app_version,
sha256 }
  content.db         SQLite (schema in 05-data-design.md section 3)
  signature.ed25519  signature over manifest.json + sha256(content.db)
```

Install flow: copy to temp -> verify signature with the content keys embedded in this build profile (public build: public key only, so internal_eval packs fail here) -> verify `sha256` -> check `format_version` and entitlement -> move into `content/` -> build remedy mapping -> register. Any failure leaves existing packs untouched.

Doctor-built packs (PACK-011) are unsigned; they load with an "Unverified" badge and are stored in `user_content.db` until exported.

5. Content pipeline (internal tool)

Diagram: diagram 15-pack-pipeline. `tools/content-builder` stages: check registry row (CONT-006) -> import (OOREP dump / CSV / OCR text / publisher XML) -> intermediate JSON -> normalise remedy abbreviations -> proofread queue (web UI, local) -> build `content.db` + FTS + `top5_json` -> sign with the channel's key (public or eval, CONT-002) -> import report (CONT-010) -> publish to that channel.

5b. Content channels (CONT-001..010)

Diagram: diagram 21-content-channels.

	Internal build profile	**Public build profile**
Who gets it	Our team, pilot testers under agreement	Doctors (public release)
Embedded content keys	public + eval	public only
Bundled packs	all internal_eval + public packs	public packs only
Eval pack copied in by hand	opens	rejected at signature check
Marking	permanent banner, "EVALUATION" on prints	none
CI gate	none	fails if any eval pack, eval key or `content-sources/internal-eval/` file is present (CONT-005)

`import-oorep` mapping (OOREP PostgreSQL dump -> pack schema, 05-data-design.md section 3):

OOREP table	**Pack table**	**Notes**
`info` (abbrev, title, languag, author, yearr, license, edition)	`pack_meta`	one pack per `abbrev`; licence copied to manifest
`chapter` (abbrev, id, textt)	`chapter`	
`rubric` (abbrev, id, mother, ismother, chapterid, fullpath, path, textt)	`rubric`	`mother` -> `parent_id`; `fullpath` -> `full_path`; depth from path
`rubricremedy` (abbrev, rubricid, remedyid, weight, chapterid)	`rubric_remedy`	`weight` 1-3 -> `grade`; grade_scale 1..3
`remedy` (id, nameabbrev, namelong, namealt)	`remedy` + remedy master mapping	PACK-008

OOREP table	Pack table	Notes
`mminfo`, `mmchapter`, `mmsection`	`mm_book`, `mm_section`	Boericke; section tree via parent/successor ids
`member`, `caze`, `cazerubric`, `file`, `activity`, `emailhistory`, `passwordchangerequest`	ignored	OOREP user/account tables, never imported

6. Citations (CIT-001..010)

- `source` rows are reusable; each installed pack auto-creates a source row.
- `citation` holds page ref, quote, confidence, optional patient/visit link.
- `citation_link(target_type, target_id)` attaches one citation to any target (rubric, rubric-remedy, user rubric, MM section, analysis rubric, prescription).
- UI quick-add: rubric panel -> "Add remedy" -> remedy, grade, source (last used pre-selected), page -> Save; single transaction writes `user_rubric_remedy` + `citation` + `citation_link`.

7. UI layout (repertory workspace)

Wireframe: diagram 16-wireframe-repertory. Three panes: left = search + tree, centre = rubric detail with grade-styled remedies, right = case clipboard with weights; bottom drawer = result grid/list/graph with method selector. Keyboard: `/` focus search, `Enter` add rubric to clipboard, `Ctrl+R` run, `1-4` set weight.

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04. Licensing and Payments Design

Requirements: LIC-*, *PAY-*, LSRV-* in `requirements/03-licensing-upi.md`. MVP1 diagrams: diagram 19-seq-mvp1-licence-issue, diagram 20-wireframe-buy-screen, diagram 12-state-licence. MVP2 diagrams: diagram 09-seq-offline-activation, diagram 10-seq-upi-purchase.

1. MVP1 flow at a glance

Step	Who	Where	What happens
1	Doctor	App Buy screen (offline)	Picks 6 months / 1 year, optional offer code, confirms name, registration number, email, mobile
2	Doctor	Phone	Scans Pay QR (amount and install_id prefilled), pays with UPI PIN

Step	Who	Where	What happens
3	Doctor	Phone	Scans WhatsApp QR (message pre-typed), attaches payment screenshot, sends
4	Operator	Business UPI dashboard	Finds payment by UTR / install_id, confirms amount and date
5	Operator	Vendor Licence Tool (offline laptop)	Pastes message, enters UTR; tool checks duplicate UTR, derives plan and validity, signs licence
6	Operator	Email + WhatsApp	Sends `.hclic`, Licence Certificate PDF and invoice; WhatsApp reply "sent to your email"
7	Doctor	App (offline)	Imports `.hclic` (drag-drop / double-click), or scans certificate QR, or types key

Target: steps 4-6 within 1 hour in support hours (LIC-049). Trial or current licence keeps working throughout (PAY-016).

2. Offline Pay QR (PAY-011, PAY-012)

```
upi://pay?pa=<business UPI ID>&pn=<business name>&am=3999.00&cu=INR&tn=HC-7K3P9%201Y&tr=HC-7K3P9
```

- Built by `hc-licence::payqr` from constants stored encrypted in the binary (PAY-015) and the plan price table compiled into the release. QR rendered locally (error correction level M, size fits 1366x768 screens).
- Offer code: base32 string = {code id, sku, amount_paise, valid_until} + Ed25519 signature from a separate offer-signing key. Verified offline; replaces `am`.
- Matching keys: `tn` / `tr` carry install_id; UTR is the authoritative match (some apps drop notes).
- Fallback text with Copy buttons; payee-name check hint.

3. WhatsApp send-proof QR (PAY-013)

`https://wa.me/91XXXXXXXXXX?text=` + URL-encoded message:

```
HomeoClinic licence request
Plan: 1 year (Rs 3,999)
Name: Dr <full name>
Reg no: <number> (<council>)
Email: <email>
Mobile: <mobile>
Install ID: HC-7K3P9
```

```
App: 1.0.3
Please find payment screenshot attached.
```

The Vendor Licence Tool parses this exact format (labels are fixed, values after the colon).

4. Licence formats

Full licence (`.hclic` file and certificate QR), CBOR payload + signature, base64url:

```
{
  "v": 1, "key_id": "k1",
  "licence_id": "L-2611-000123",
  "sku": "CLINIC_1Y",
  "holder": { "name": "DR ASHA KULKARNI", "reg_no": "12345", "council": "MCH", "email":
"...", "mobile": "+91..." },
  "install_id": "HC-7K3P9",
  "issued_at": "2026-11-01", "valid_from": "2026-11-01", "valid_until": "2027-10-31",
  "entitlements": ["core"],
  "utr_hash": "sha256:..."
}
```

Typed key (LIC-046): 22-byte compact payload (licence_id, sku, valid_from/until as day numbers, 8-byte hash of normalised name + reg no, key_id) + 64-byte signature = 86 bytes -> 138 base32 data chars, split into 35 groups of 4 data chars + 1 check char = 175 characters shown as 35 groups of 5. Normalisation: uppercase, collapse spaces, strip "DR."/punctuation.

Verification on import and on each check site (TAM-004): signature with embedded key, `valid_from <= today_guarded <= valid_until (+15 d grace)`, identity hash matches the locked profile.

5. Identity binding (LIC-006)

- On first activation, `doctor.name` and `doctor.reg_no` are overwritten with licence values and marked locked; print templates read the identity only from the verified licence, not from editable settings, so editing the database does not change prints.
- Re-import on another PC: same licence, same identity, no limit. A different doctor's licence on the same install is rejected unless the profile is reset (which starts a new clinic profile, MVP1 = one licensed doctor per install).

6. Licence states

Diagram: diagram 12-state-licence.

`Unlicensed -> Trial(30d) -> Active -> Grace(15d) -> ReadOnly`, `Any -> Tampered` (signature fail confirmed twice, treated as ReadOnly with banner and Repair). `Active -> Revoked` exists only from MVP2 (online refresh). ReadOnly always allows view, search, print and export.

Clock guard: `state.bin` stores max seen UTC time, HMAC-ed with a key derived from the DEK. If now < max_seen - 48 h, use max_seen for expiry checks (LIC-005).

7. Vendor Licence Tool (LIC-040..050)

Part	Design
Platform	Same Tauri + Rust stack, internal build, runs on an offline operator laptop
Signing	Ed25519 key on a hardware security key (PIV/OpenPGP card); separate offer-code key; sealed paper backup
Input	Paste WhatsApp message (parser for section 3 format) + UTR + amount + date from UPI dashboard
Checks	Duplicate UTR (unique index), amount to plan mapping incl. offer codes, under/over payment handling, renewal stacking from current expiry
Output	`.hclic`, Licence Certificate PDF (name, reg no, plan, validity, licence id, QR, key text), GST invoice PDF, mailto/SMTP draft with attachments, WhatsApp reply text
Register	SQLCipher DB: customer, licence, payment(utr unique), invoice, offer_code, dealer, audit; Excel export; daily encrypted backup; MVP2 import format

8. Backend services (MVP2)

Endpoint	Auth	Purpose
`POST /v1/activate`	licence key	Online activation
`POST /v1/activate/offline`	portal session (OTP)	Request code -> response code
`POST /v1/deactivate`	signed deactivation code	Free a device seat
`GET /v1/licence/refresh`	device token	Renewals, revocation
`POST /v1/orders`	portal session	Create Razorpay order / subscription
`POST /v1/payments/webhook`	Razorpay HMAC signature	Only source of truth for payment
`/portal/*`	phone OTP session	Customer self-service
`/admin/*`	SSO + 2FA, IP allowlist	Issue/extend/revoke, refunds, dealers

Tables: `customer`, `order`, `payment`, `subscription`, `licence`, `device`, `invoice`, `dealer`, `audit`. Webhook handler is idempotent on `razorpay_payment_id` (unique index) and issues the licence in the same transaction as marking the order paid.

9. Hosted UPI flows (MVP2, PAY-002, PAY-006)

- One-time purchase: Razorpay Order -> Checkout with UPI Intent (mobile) or UPI QR (desktop browser) -> webhook `payment.captured` -> licence issued -> SMS + email with key, response code and GST invoice.
- Subscription: Razorpay Subscription with UPI AutoPay mandate -> webhook `subscription.charged` each cycle extends `valid_until` -> desktop picks up on next online refresh, or doctor imports new response code. Pre-debit notification handled by Razorpay per RBI rules.

10. Crack resistance of the licence path (TAM)

- `hc-licence` is built with control-flow obfuscation (TAM-008); its public key and messages are compile-time encrypted (TAM-003).
- Verification is not one function call. Four independent check sites (startup, write commands, pack open, 20-60 min random timer) each re-verify the signature and derive the licence state differently, and the results feed the encryption key path of paid packs, so a patched "return true" still cannot decrypt paid content (TAM-004, TAM-009).
- MVP1: no paid packs; sharing is deterred by identity binding (section 5, TAM-014) and forging is impossible without our signing key.
- MVP2 paid pack key flow: server generates `pack_key` once per pack; for each licence it adds `wrapped_key = AES-KW(KDF(licence_secret, device_fp), pack_key)` to the signed licence (LIC-031). The app unwraps in memory at pack open and zeroises after use.
- Watermark: `licence_id` is folded into the per-licence wrapped keys and into invisible ordering markers in paid pack exports, so a leaked pack or key traces back to the buyer (TAM-010).
- MVP2 server signals: activation count per key, fingerprint churn, geographic spread, refresh from known cracked builds (integrity report from TAM-005) -> automatic revocation and update block (TAM-011).

---

05. Data Design

Requirements: `requirements/06-architecture-data-model.md` sections 2-4b (schemas are defined there and not repeated here). This document adds the design decisions, indexes, backup format and legacy import. Diagrams: diagram 05-er-clinic, diagram 06-er-content, diagram 07-seq-patient-search, diagram 11-seq-backup-restore.

1. Conventions

Item	Rule
Keys	UUIDv7 text (sortable by time), patient `code` is the human ID
Money	INTEGER paise
Time	UTC ISO-8601 text; display in Asia/Kolkata
Soft delete	`deleted_at`; all queries filter it via views `v_patient`, `v_visit`
JSON	`data_json`, `vitals_json` validated against the template JSON schema in Rust before write
Text	UTF-8, NFC normalised

2. Patient search storage (PSR, NFR-002)

Diagram: diagram 07-seq-patient-search. - `patient_fts` (unicode61, prefix 1-3) and `patient_trgm` (trigram on digits) as external-content FTS5 tables, kept in sync by triggers in the same transaction (PSR-012). - Denormalised summary columns on `patient`: `last_visit_at`, `next_followup_on`, `balance_paise`, `visit_count`. - Sizing at 50k patients / 500k visits: clinic.db about 400-700 MB without attachments; FTS indexes about 15 MB.

3. Content pack DB

Schema per requirements 06 section 4, plus `rubric.top5_json` precomputed and `remedy_index` (global position for bitsets). Measured from the OOREP dump (2026-10-15): publicum 74,667 rubrics / 735,566 remedy entries, kent-de 68,741 / 624,010, remedy master 2,432. Estimated pack size 40-60 MB each including FTS; to be confirmed when M4 builds the packs.

4. Migrations

`hc-storage/migrations/NNNN_name.sql`, forward-only, applied in one transaction after automatic pre-migration backup (NFR-011). Schema version in `PRAGMA user_version`. CI test migrates a DB from every released version.

5. Backup format `.hcbak` (BAK-002)

```
magic "HCBAK1" | header JSON (created_at, app_version, patient_count, last_visit_at, kdf
params, salt, nonce)
| AES-256-GCM( tar( clinic.db snapshot, user_content.db snapshot, files/* ) ) streamed in 1
MB chunks, each chunk authenticated
```

Snapshot via SQLite online backup API, so backups run while the app is in use. Restore preview reads only the header (BAK-003).

6. Legacy import (`tools/legacy-import`, also an in-app wizard) - MVP5, de-prioritised

Not built for MVP1. Design kept ready for IMP-002.

Source: live legacy MySQL or a `.sql` dump (parsed without needing MySQL installed).

Legacy table	New target	Notes
tblpatient	patient	CaseNo -> `code` with prefix kept; age-only -> `dob_is_estimated`
tblchiefcomplaint, tblhistory, tblgeneralities, tblmentals, tblmenstrualhistory	first `visit` (type First) `data_json` using the "Legacy case" template	Field-by-field mapping table in the tool
tblfollowup	visit (type Follow-up)	`Rubrics` text kept as note; optional fuzzy match to Kent rubrics offered for review
tblpriscription	prescription + prescription_item	Medicine text parsed to remedy + potency when confident, else kept as label
paymentcustomer, tblreceipt	invoice, invoice_line, payment	double -> paise rounding
tblappointment	appointment	
tbldiet, tblinstruction	advice library	Shivaji-font Marathi text converted to Unicode Devanagari using a glyph mapping table
tblreports	attachment	Files copied and encrypted
Patient Photos/	attachment (photo)	Matched by case number in file name
tblclinicsetting	clinic, doctor	
login, tbllicenceinfo	not imported	New PIN and new licence

Import is dry-run first (report of counts, unmapped values, conversion warnings), then committed in one transaction.

---

06. Review Questions and Proposed Changes

Please mark each item Accept / Reject / Change in the review.

Proposed requirement additions (from legacy analysis)

Q	Proposal	Reason
Q1	Add DOC-001..005 to MVP1 as P1: medical certificate, referral letter, envelope print, patient declaration, birthday reminder list	Legacy users print these today (Prints/Certificate, Envelope, Letter, Declaration; BirthdayReminder class). Dropping them would block migration.
Q2	RESOLVED: legacy Doctor Clinic importer moved to MVP5 (IMP-002) with IMP-001; design kept ready	Owner decision 2026-10-14: focus on complete offline MVP1 first
Q3	Add Marathi advice/diet print in MVP1 (UI stays English)	Legacy prints Marathi instruction and diet sheets; current spec puts languages in MVP2.

Owner decisions (2026-10-14)

- 2026-10-15: MVP1 integrates every repertory/MM we can obtain (OOREP publicum, kent-de, Boericke) as plug-and-play internal-eval packs for testing only; they are excluded from public builds automatically (requirements 02 sections 2b-2c, CONT-001..010).
- MVP1 is a complete offline product with no hosted servers; payment by offline-generated UPI QR to our business UPI ID, proof via pre-typed WhatsApp QR with screenshot, signed licence + certificate PDF emailed from the offline Vendor Licence Tool, doctor imports it offline (requirements 03 section 3).
- Products for MVP1: 6-month and 1-year licences only.
- AWS backend, portal, cloud backup and mobile apps start from MVP2 only once MVP1 revenue funds them; designs stay ready.
- Migration (legacy and competitor import) is MVP5.

Defaults chosen for licensing (please confirm or change)

Q	Default used in requirements	Alternative
Q11	Licence certificate PDF is not password protected; the Ed25519 signature makes it tamper-proof (LIC-045)	Password = registered mobile number

Q	Default used in requirements	Alternative
Q12	Licensed name + registration number are locked and printed on all documents (LIC-006); second doctor needs own licence (LIC-007)	Allow a named assistant doctor on the same licence
Q13	No PC limit: unlimited reinstalls for the same doctor (LIC-064); install_id recorded only	Cap at 2 PCs using install_id or fingerprints (needs re-issue on PC change)
Q14	Placeholder prices Rs 2,299 / 6 months and Rs 3,999 / year; UPI only, no cards in MVP1	Final prices from you
Q15	Support hours Mon-Sat 10:00-19:00 IST, licence within 1 hour (LIC-049)	Your actual hours

Content questions

Q	Question	Proposed default
Q16	Ask OOREP author (Andreas Bauer) for permission to use his English Kent, Boger, Boenninghausen and Boericke conversions in a closed-source product?	Yes, ask now; it could remove months of OCR work
Q17	Who may receive internal builds (team only, or also pilot doctors under a written evaluation agreement)?	Team only until legal review of GPL content

Design decisions needing confirmation

Q	Question	Proposed default
Q4	Tauri 2 + React + Rust (ADR-001, 003) instead of staying on Java	Tauri; legacy Java code has no source and is not reusable
Q5	Rx printing from HTML templates via webview (ADR-008) replacing Jasper	HTML templates; doctor can adjust letterhead margins
Q6	Backend on AWS ap-south-1 (ECS Fargate + RDS + KMS) from MVP2	Yes when funded; start MVP2 on a single small VM if cost matters
Q7	Build plan of about 3.5 months with 3 developers + 1 content person (diagram 17)	Indicative; Kent digitisation (M4) is the critical path
Q8	Is the legacy "Doctor Clinic" our own product (so its users and data format are ours to migrate), or a third-party product used as reference?	Assumed ours

Q	Question	Proposed default
Q9	Keep a single PIN per clinic in MVP1 (no separate receptionist login until MVP2)	Yes

Risks

Risk	Impact	Mitigation
Kent digitisation quality	Wrong remedies in results	Proofreading targets, golden checks against printed book samples
Low-end Windows PCs without WebView2	App will not start	Bundle offline WebView2 installer (NFR-040)
Doctors forget backup password	Backup unusable	Printed recovery key at setup, local backups also keyed by DEK
Licence cracking	Revenue loss	Layered anti-tamper (02 section 6b, TAM-001..013): hardened single binary, integrity checks, many licence checks, paid packs encrypted with device-bound keys, watermark tracing. Cannot be made 100% uncrackable, but a cracked copy gets no paid content, updates or cloud
Eval content leaks into a public release	GPL / copyright exposure	Separate eval signing key, public build cannot open eval packs, CI release gate (CONT-003, CONT-005), registry with permission status
No public repertory ready at launch	Public MVP1 has nothing useful to repertorise	Start OCR of public-domain Kent early (M4 critical path) and ask OOREP author for permission in parallel (Q16)
Anti-tamper false positives	Genuine doctor locked out or antivirus flags app	Only Read-only response (TAM-013), no packers, AV scan on every release, reinstall/online re-check clears it
Security slows or annoys doctors	Doctors abandon the app	SUX-001..012 as release gates: budgets measured on protected build, no dialogs, offline forever, two-check confirmation, self-repair, never interrupt a visit

Added question

Q	Question	Proposed default
Q10	Budget for an EV code-signing certificate (about Rs 25-40k/year) and an optional commercial obfuscator for the licence module	EV cert yes (also removes SmartScreen warnings); start with open-source obfuscation, buy commercial only if cracks appear

Diagrams

Figure 01: System context (source: diagrams/src/01-system-context.puml)

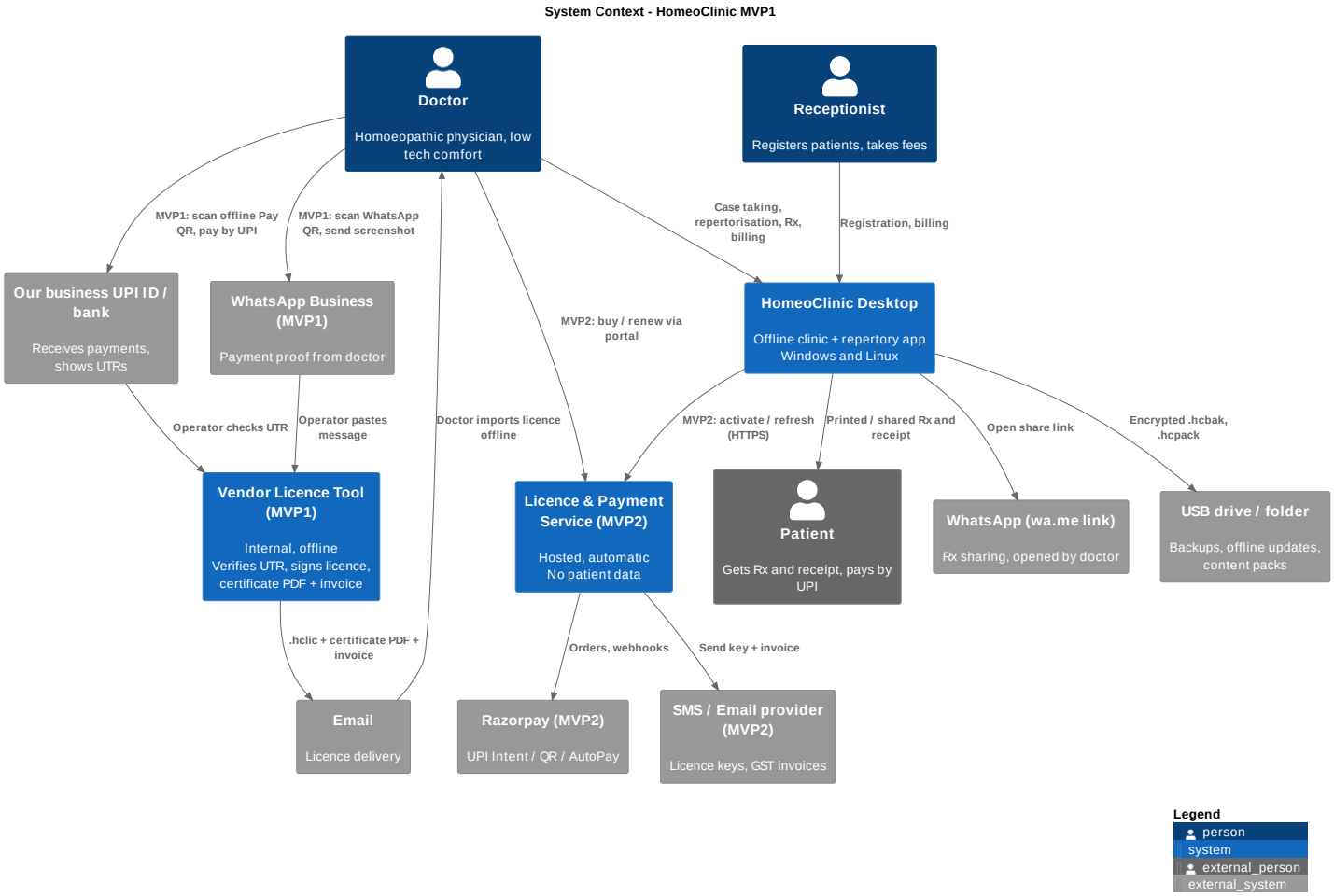


Figure 02: Containers (source: diagrams/src/02-containers.puml)

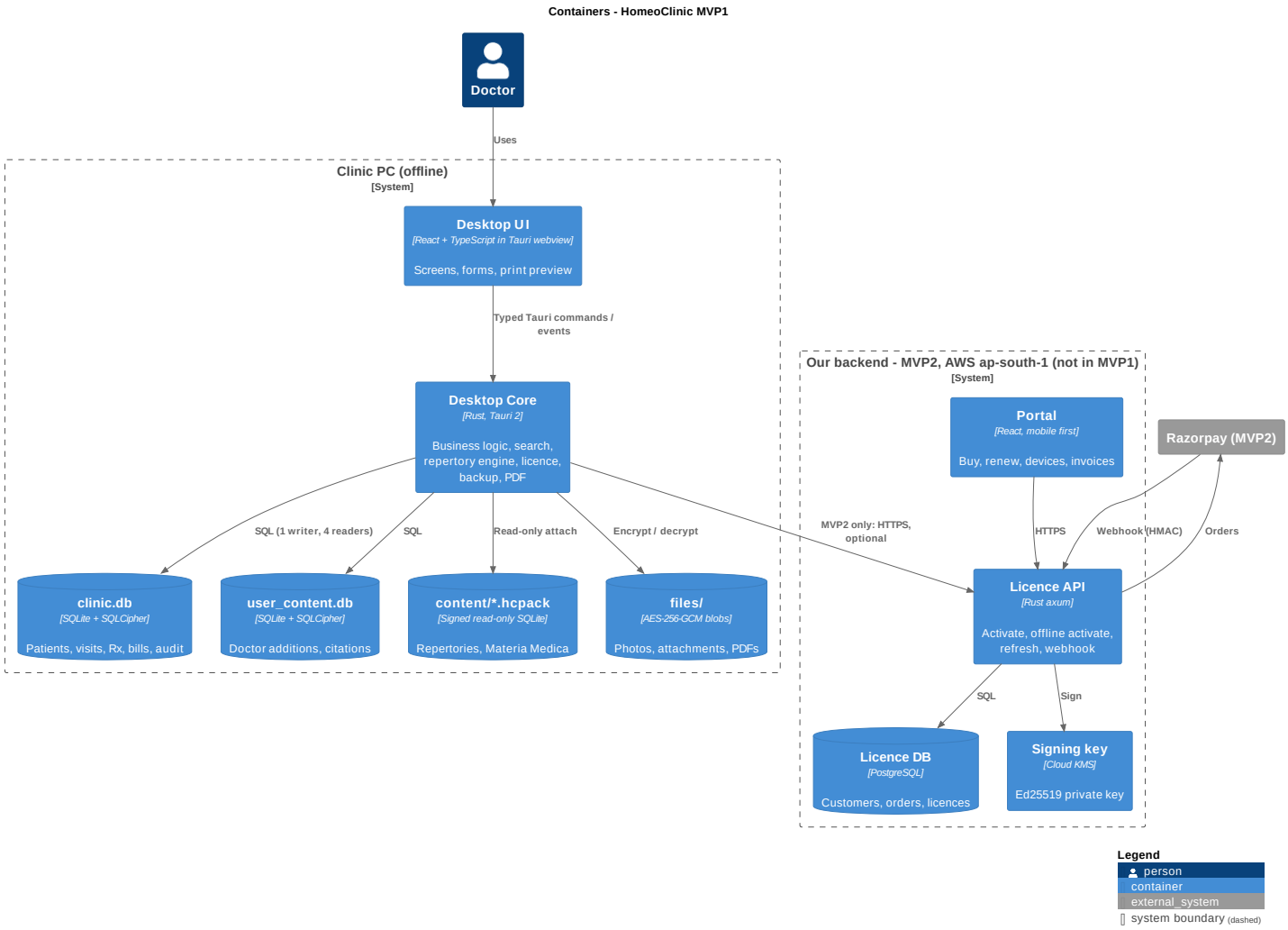


Figure 03: Desktop components (source: diagrams/src/03-desktop-components.puml)

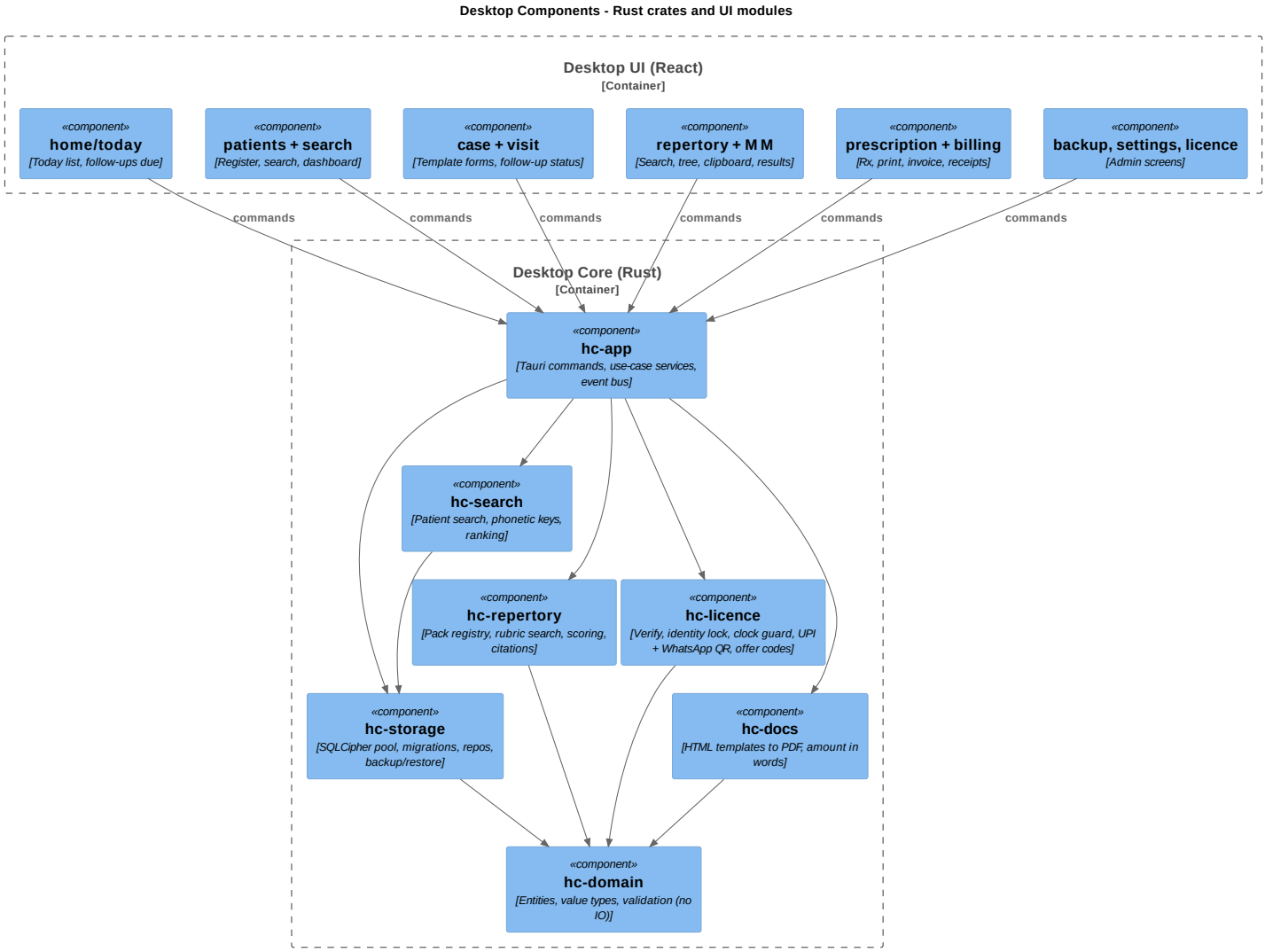


Figure 04: Deployment (source: diagrams/src/04-deployment.puml)

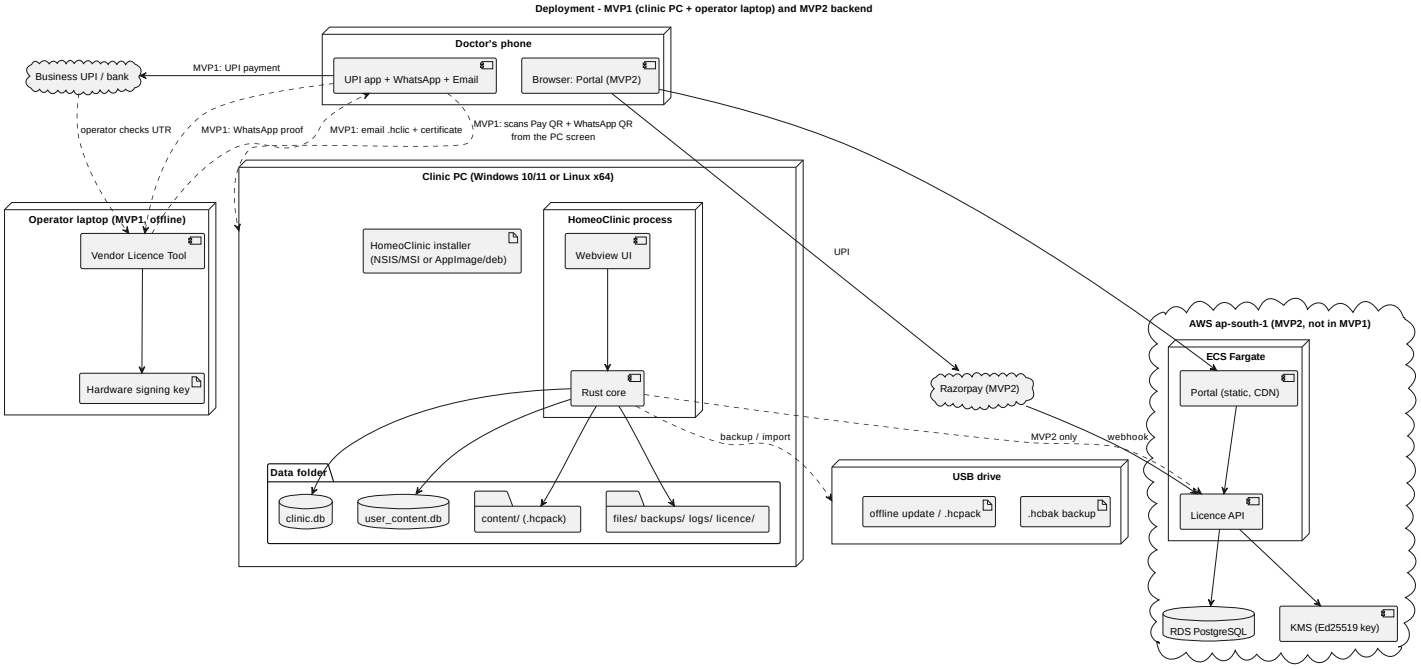


Figure 05: Er clinic (source: diagrams/src/05-er-clinic.puml)

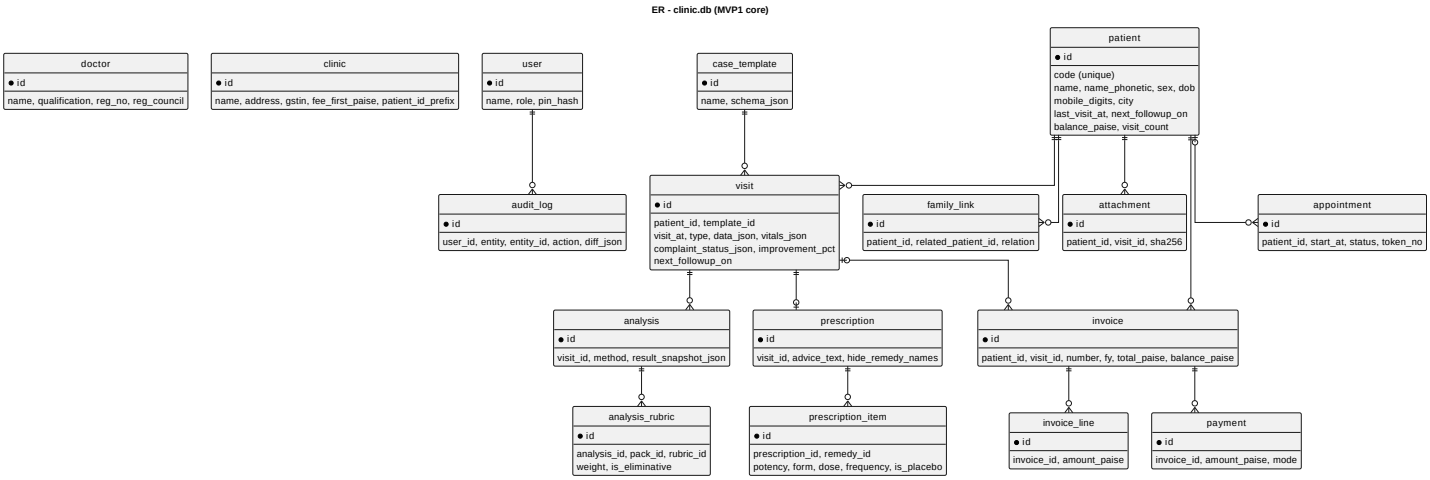


Figure 06: Er content (source: diagrams/src/06-er-content.puml)

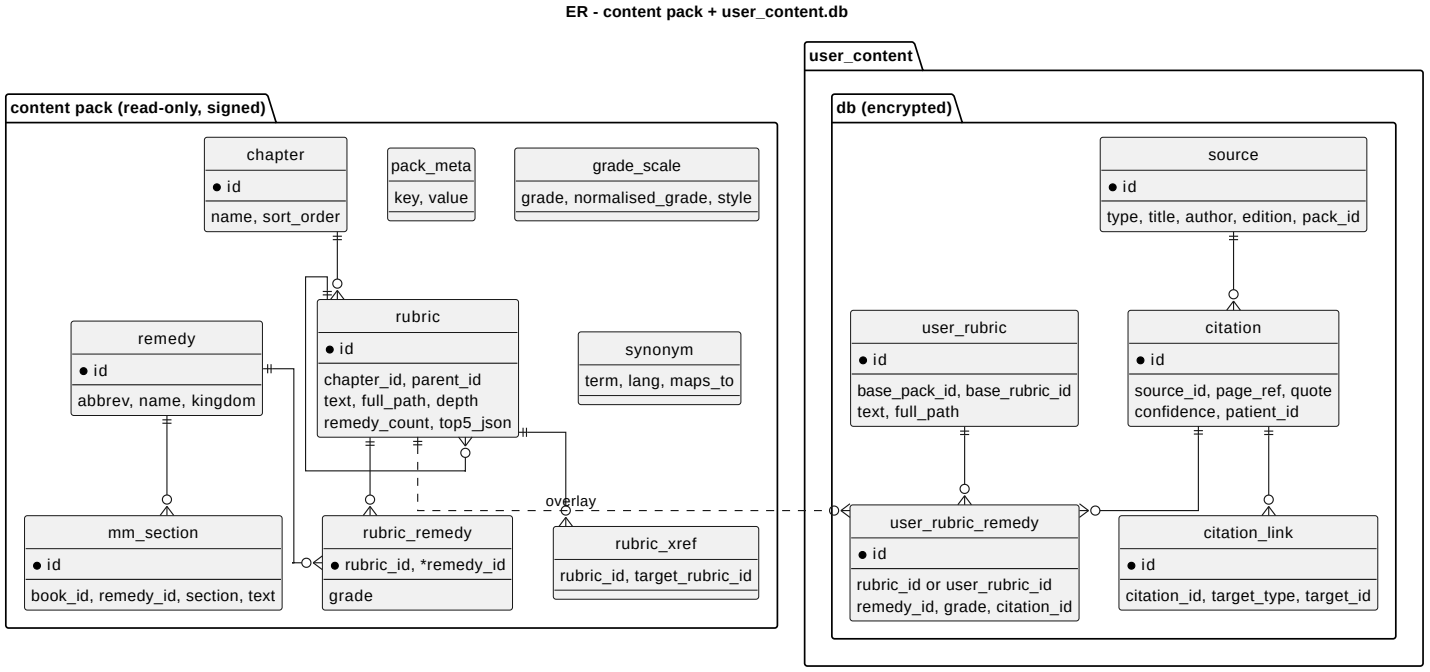


Figure 07: Seq patient search (source: diagrams/src/07-seq-patient-search.puml)

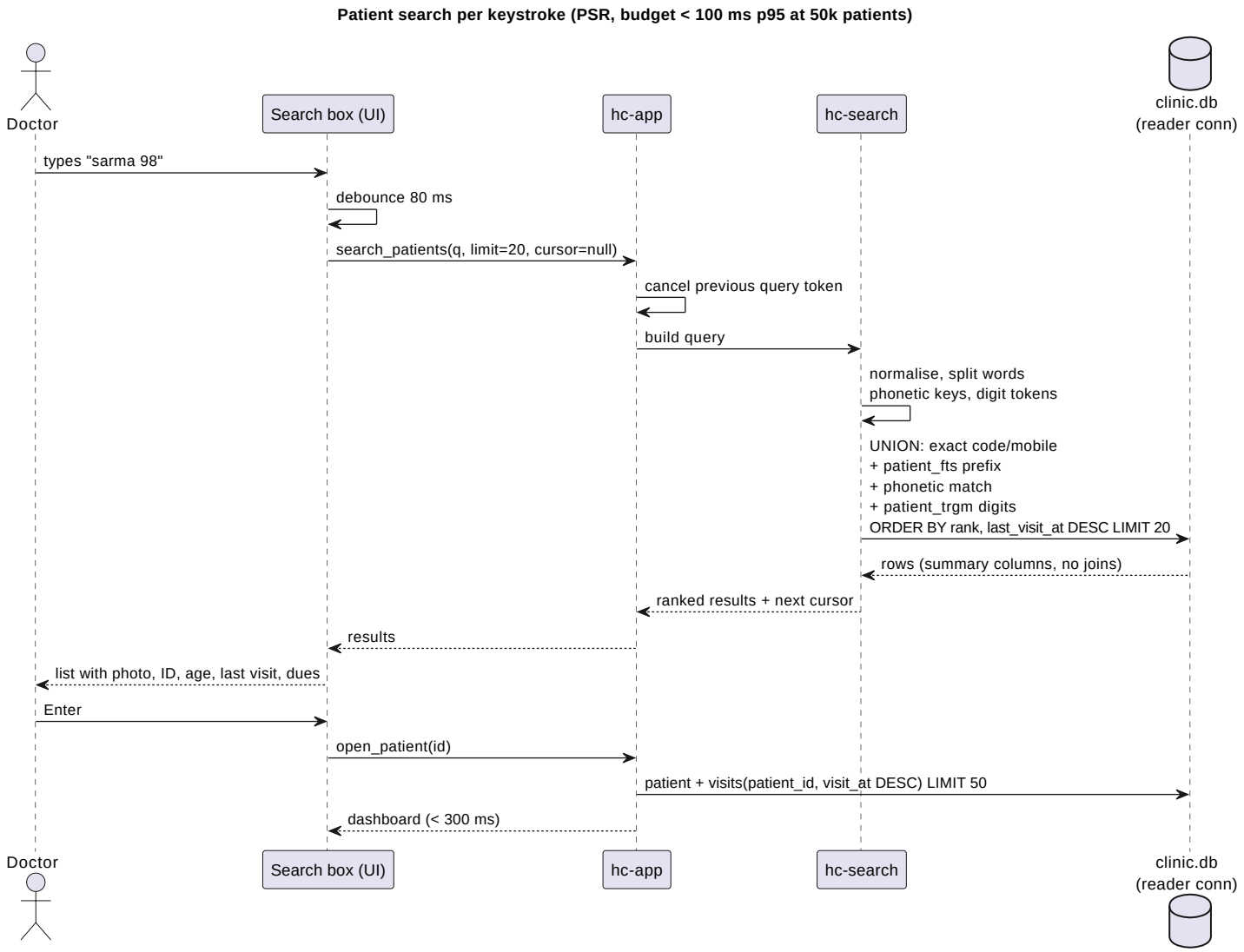


Figure 08: Seq repertorise (source: diagrams/src/08-seq-repertorise.puml)

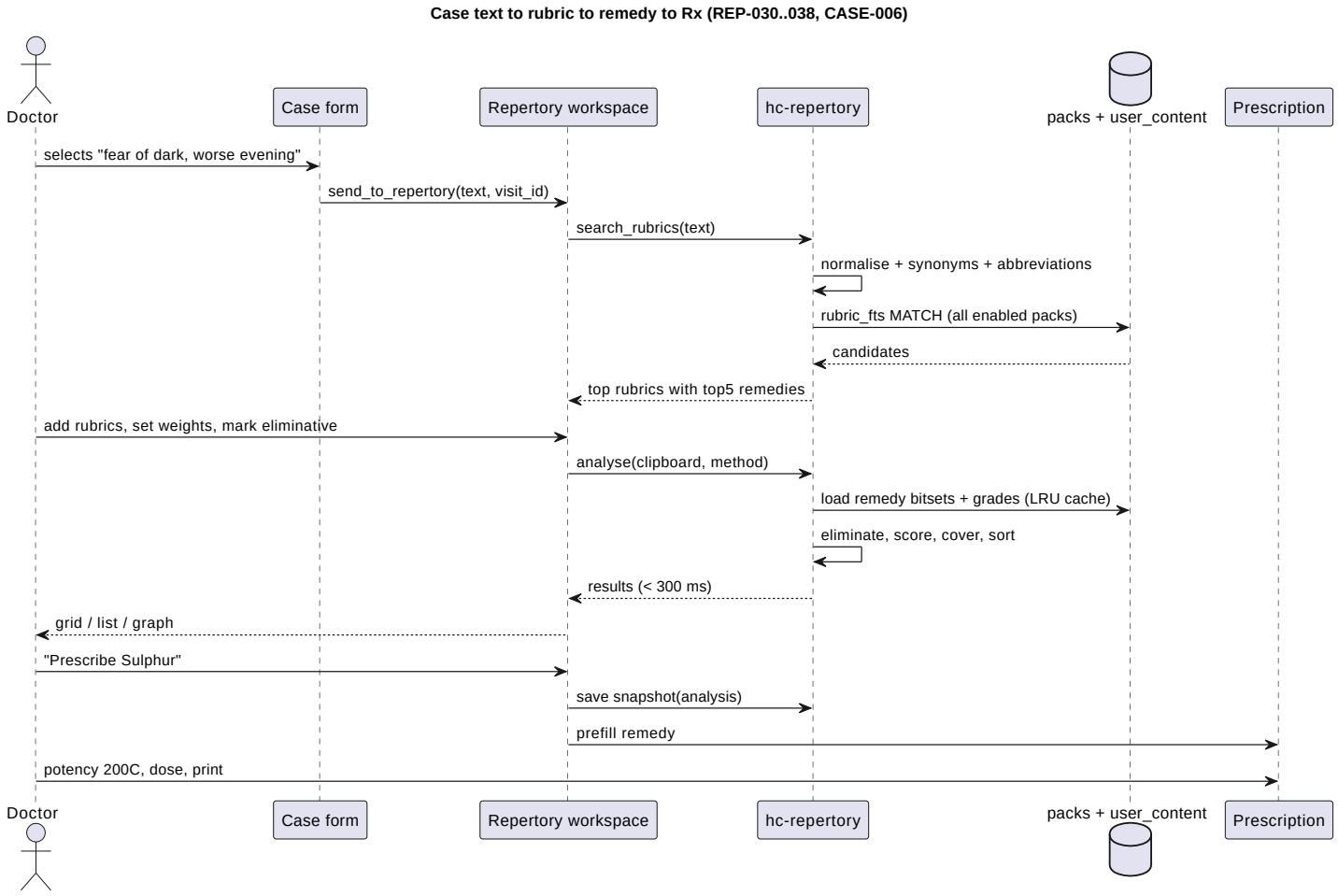


Figure 09: Seq offline activation (source: diagrams/src/09-seq-offline-activation.puml)

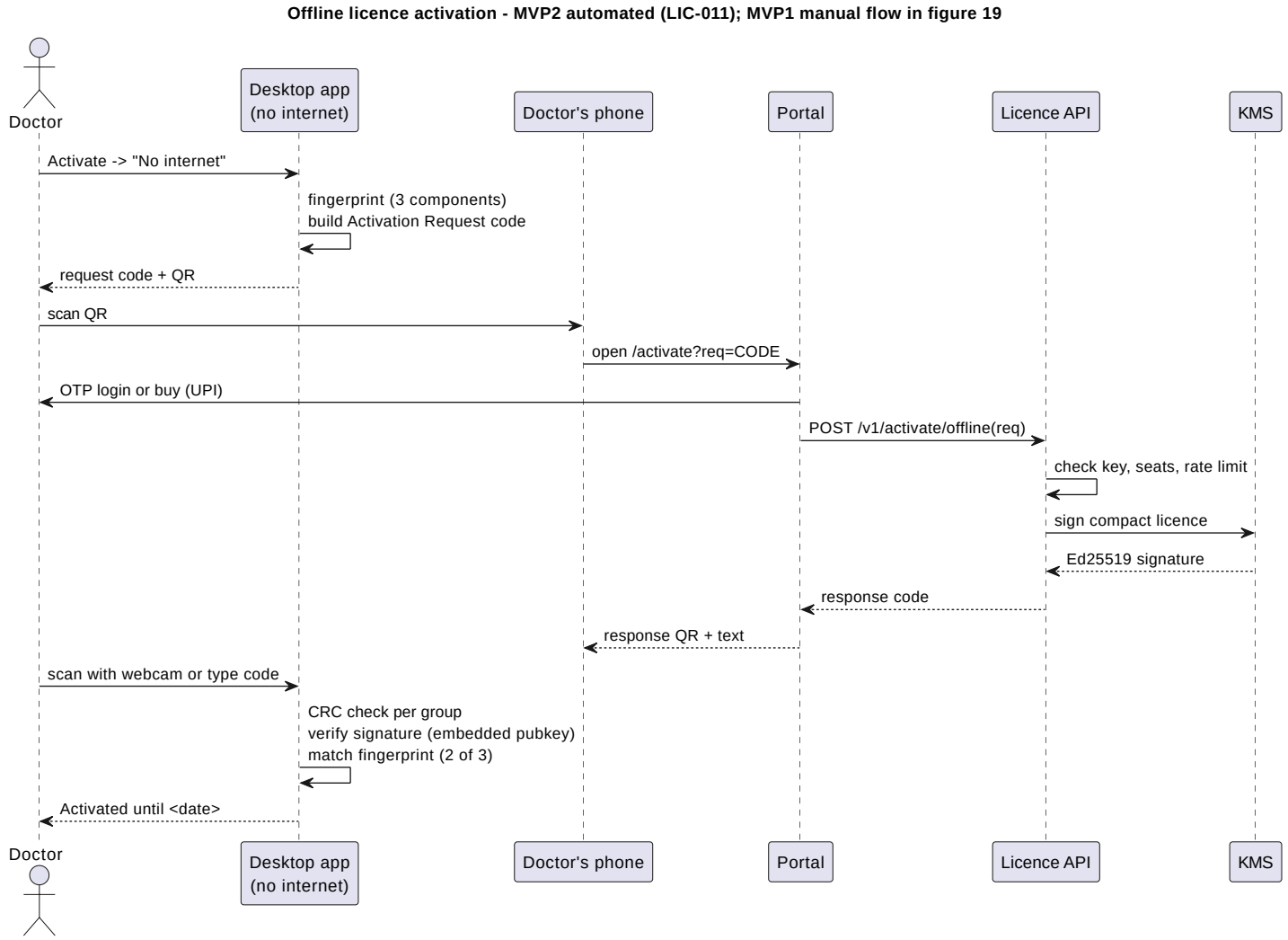


Figure 10: Seq upi purchase (source: diagrams/src/10-seq-upi-purchase.puml)

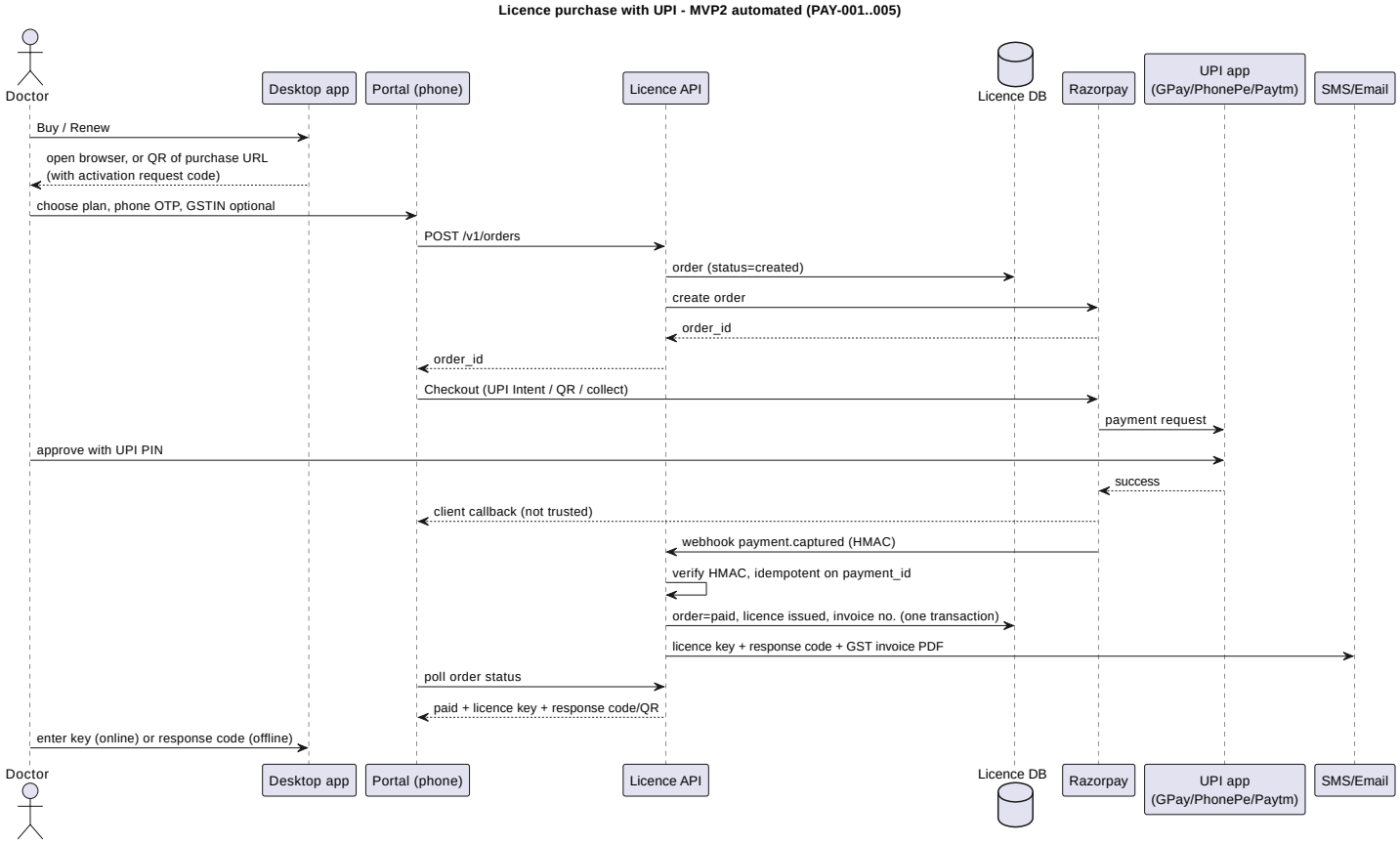


Figure 11: Seq backup restore (source: diagrams/src/11-seq-backup-restore.puml)

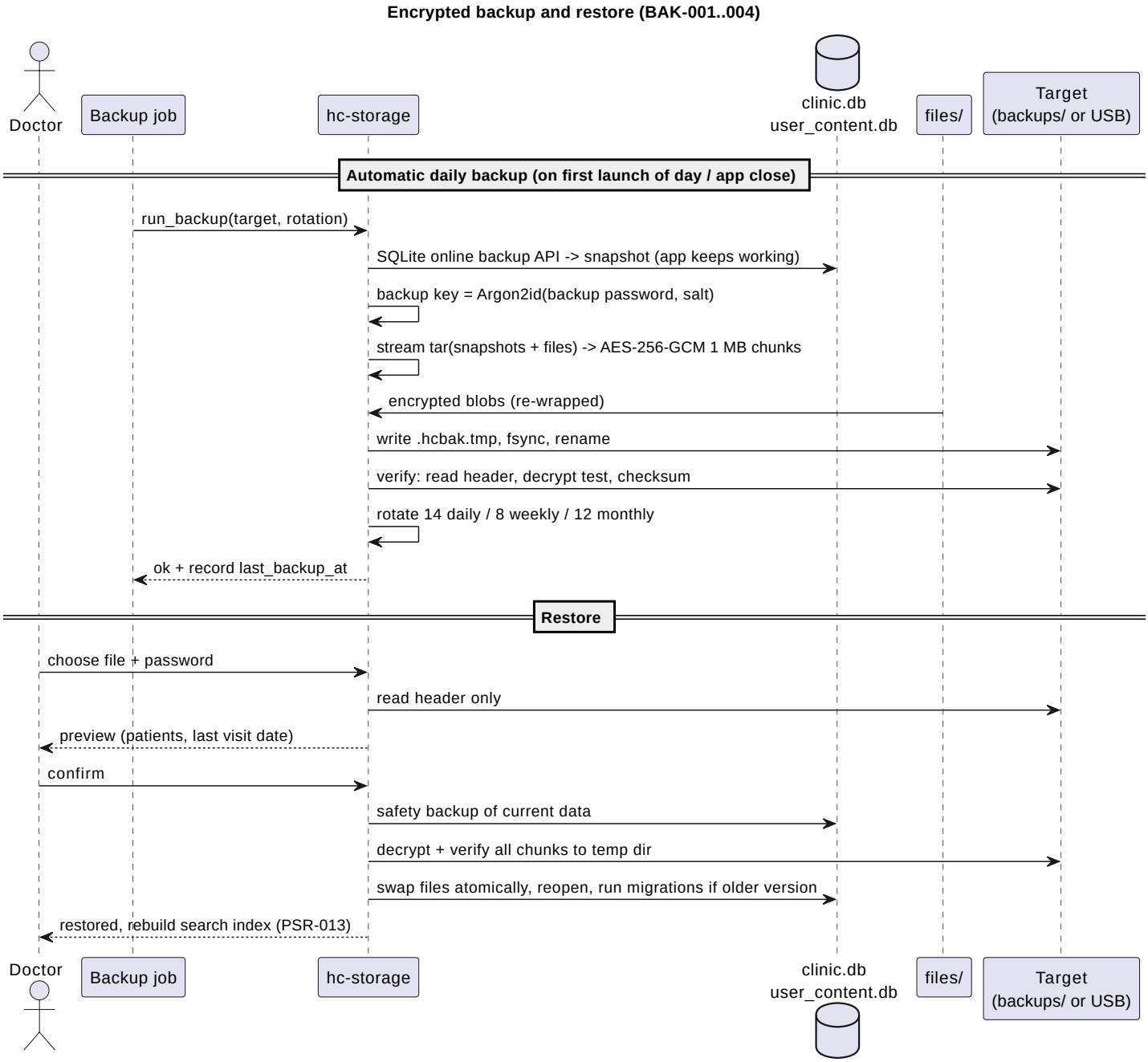


Figure 12: State licence (source: diagrams/src/12-state-licence.puml)

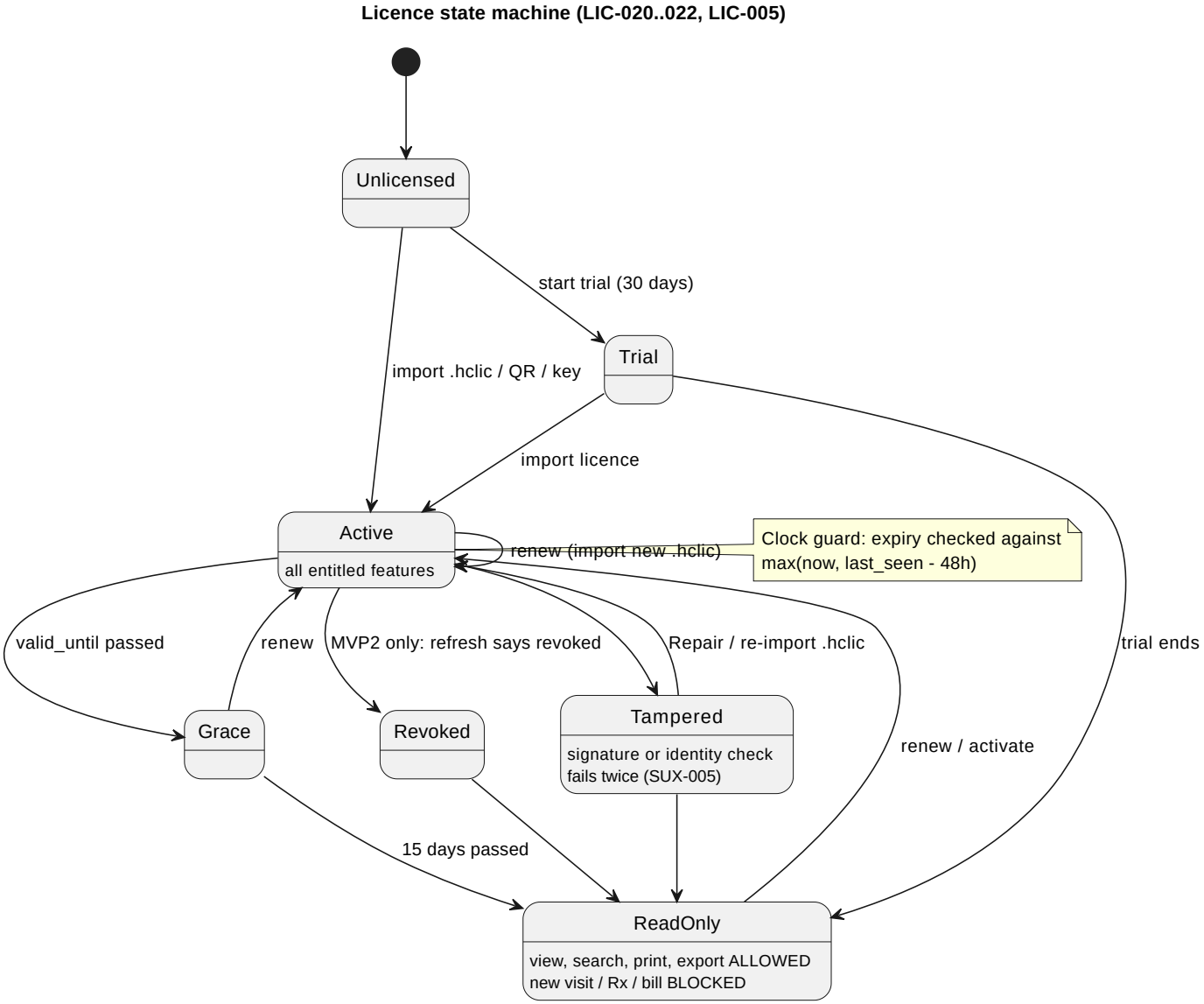


Figure 13: Legacy architecture (source: diagrams/src/13-legacy-architecture.puml)

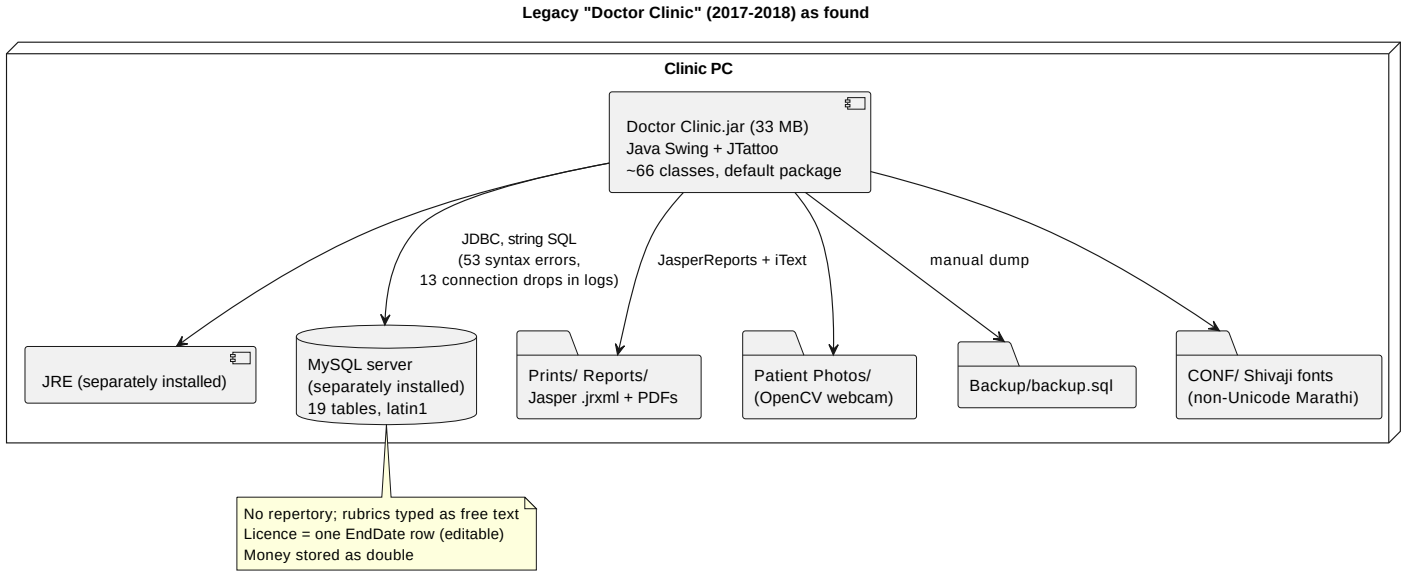


Figure 14: Seq startup (source: diagrams/src/14-seq-startup.puml)

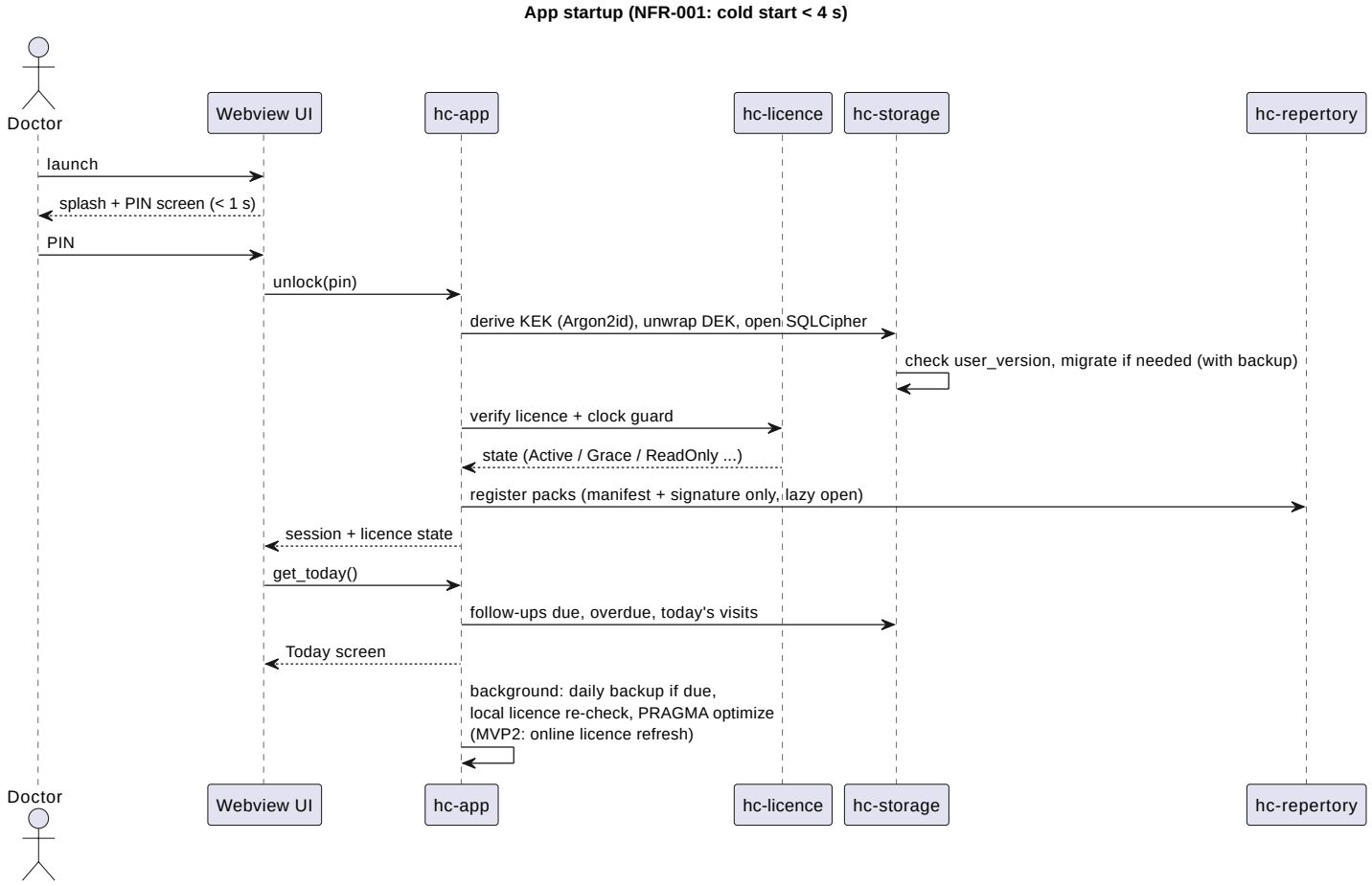


Figure 15: Pack pipeline (source: diagrams/src/15-pack-pipeline.puml)

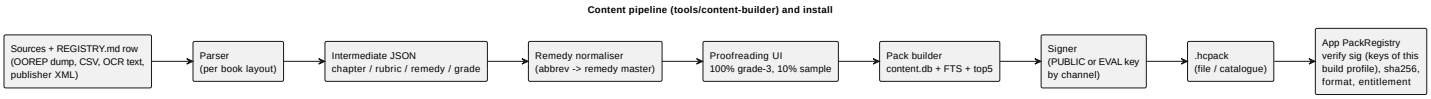


Figure 16: Wireframe repertory (source: diagrams/src/16-wireframe-repertory.puml)

Wireframe - Repertory workspace (patient: HC-000123)

<- Patient | fear dark | Search | All repertories | Run analysis

Search results

MIND - FEAR - dark, of (42)

MIND - FEAR - evening (31)

MIND - FEAR - alone, of being (88)

Chapters

+ MIND

+ HEAD

+ GENERALITIES

Rubric detail

MIND - FEAR - dark, of

CALC. CARB. PHOS. PULS. STRAM.

ars. caust. lyc. med.

acon. bar-c. ...

Cross-ref: FEAR - night

Add to clipboard

Add remedy + citation

Case clipboard

- [x] Fear dark (eliminative) weight 2
- [x] Thirstless weight 1
- [x] Chilly weight 1
- [ ] Desire sweets weight 1

Method: Totality + grade

Analysis result

(•) Grid ( ) List ( ) Graph

Remedy	Score	Coverage	Action
Puls.	14	4/4	Prescribe
Phos.	11	4/4	Prescribe
Calc.	10	3/4	Prescribe
Lyc.	8	3/4	Prescribe

Figure 17: Mvp1 milestones (source: diagrams/src/17-mvp1-milestones.puml)

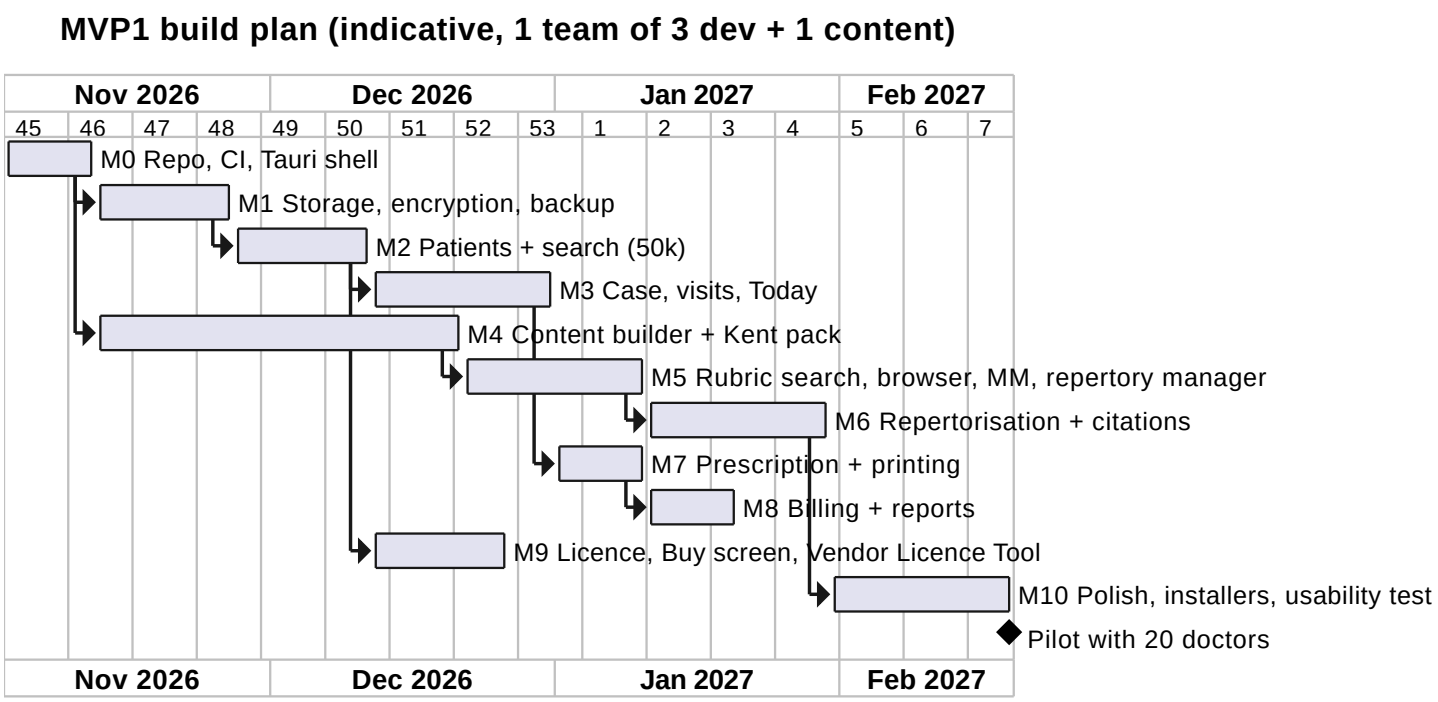
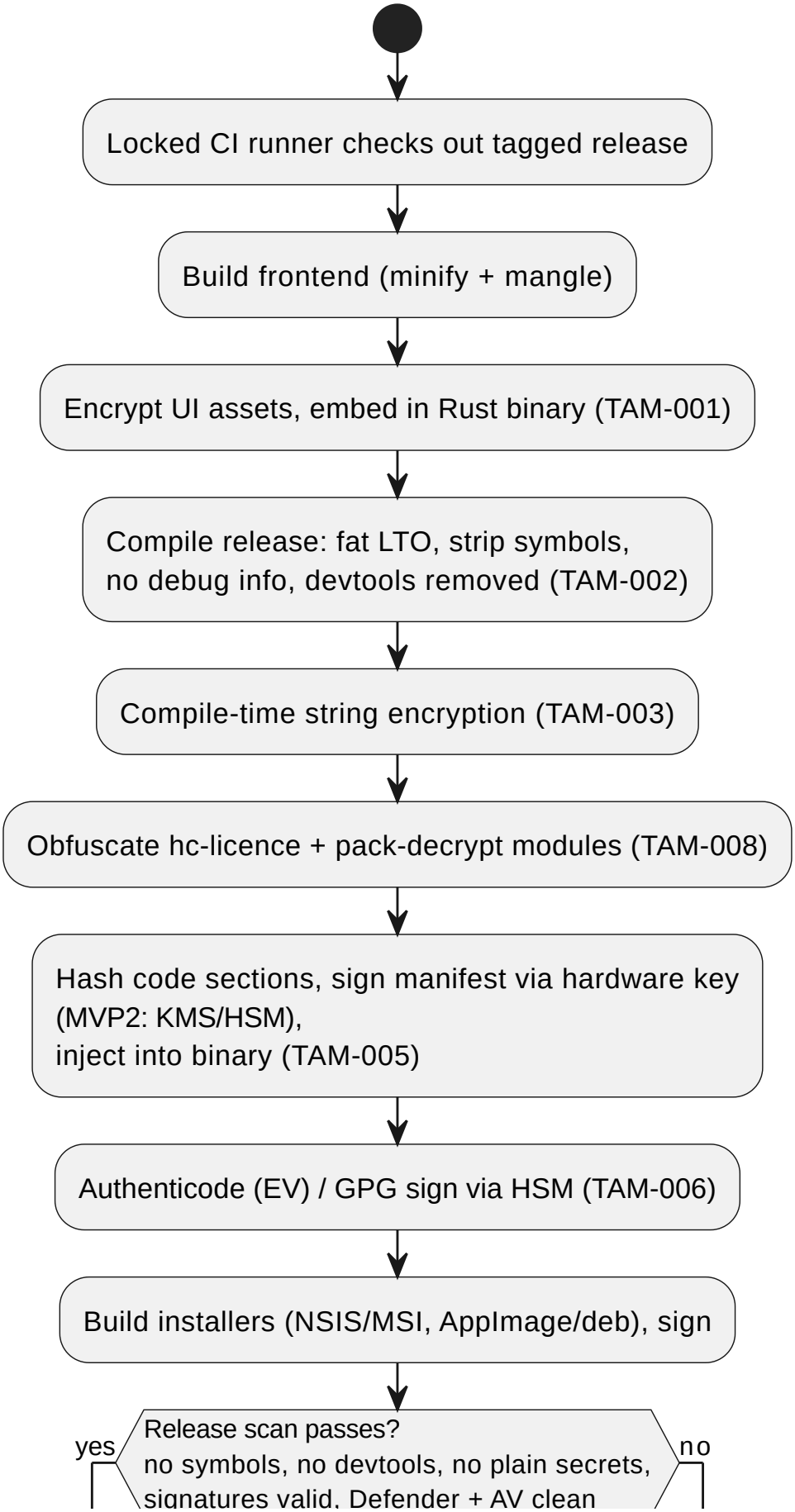


Figure 18: Secure release pipeline (source: diagrams/src/18-secure-release-pipeline.puml)

Tamper-proof release pipeline (TAM-001..012)



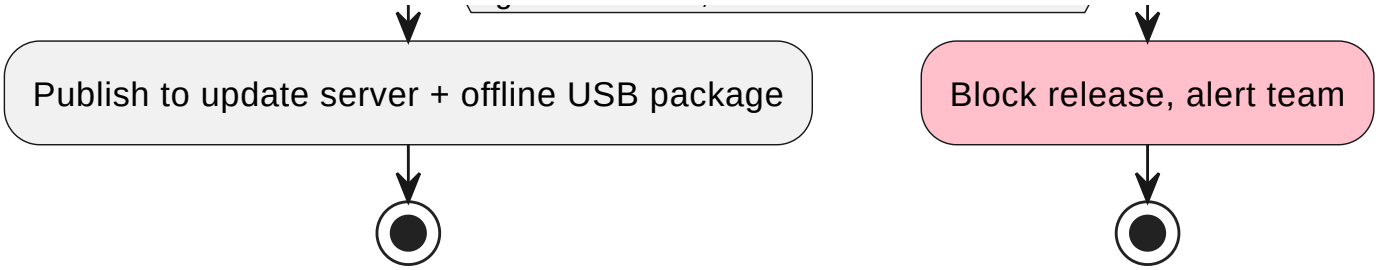


Figure 19: Seq mvp1 licence issue (source: diagrams/src/19-seq-mvp1-licence-issue.puml)

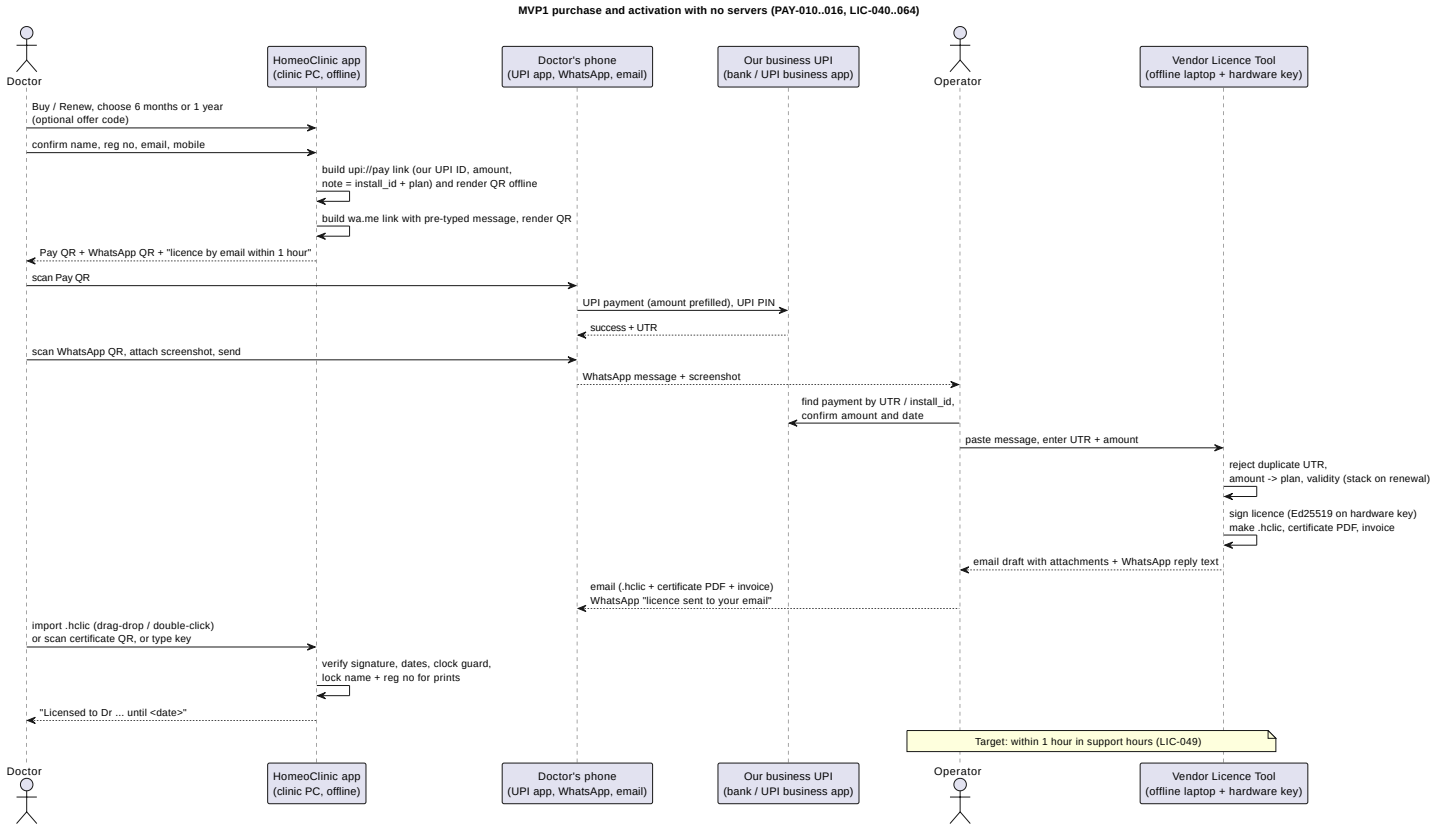


Figure 20: Wireframe buy screen (source: diagrams/src/20-wireframe-buy-screen.puml)

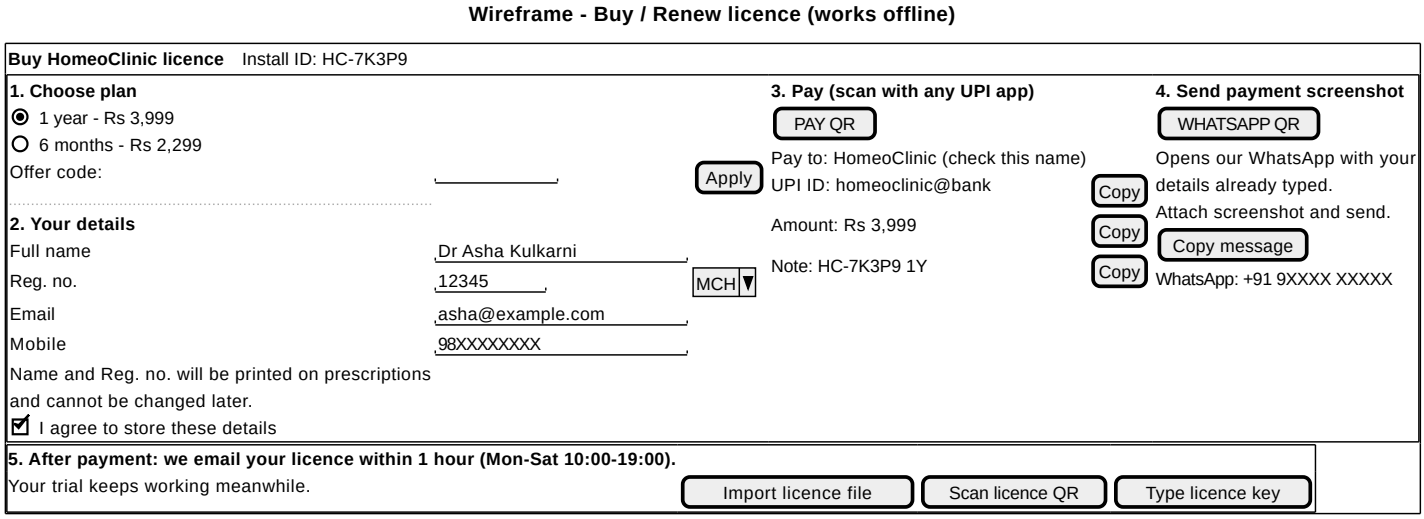


Figure 21: Content channels (source: diagrams/src/21-content-channels.puml)

